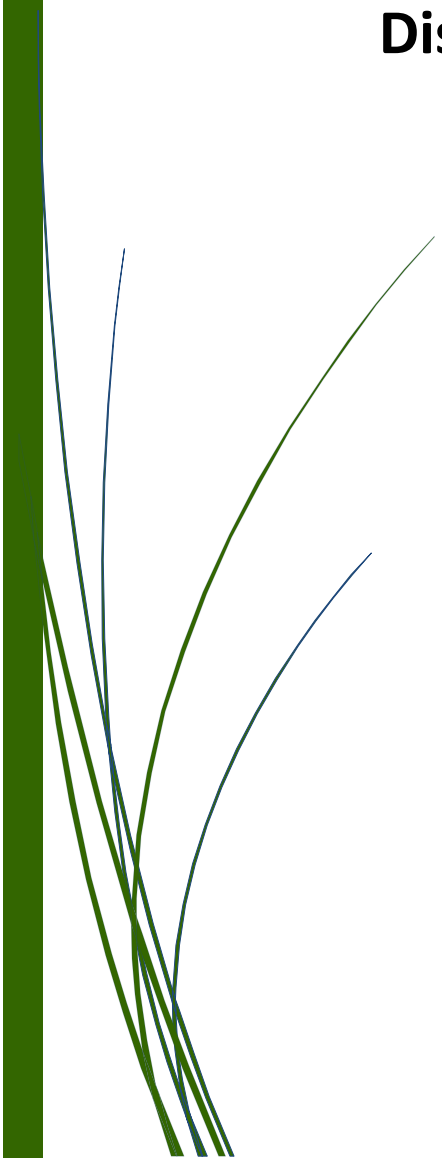
BRAC UNIVERSITY

Inspiring Excellence

Summer-2024

CSE 230

Discrete Mathematics



The Foundations: Logic and Proofs

The rules of logic specify the meaning of mathematical statements. For instance, these rules help us understand and reason with statements such as “There exists an integer that is not the sum of two squares” and “For every positive integer n, the sum of the positive integers not exceeding n is n(n + 1)／2.” Logic is the basis of all mathematical reasoning, and of all automated reasoning. It has practical applications to the design of computing machines, to the specification of systems, to artificial intelligence, to computer programming, to programming languages, and to other areas of computer science, as well as to many other fields of study.

1.1 Propositional Logic

The rules of logic give precise meaning to mathematical statements. These rules are used to distinguish between valid and invalid mathematical arguments. Because a major goal of this book is to teach the reader how to understand and how to construct correct mathematical arguments, we begin our study of discrete mathematics with an introduction to logic.

1.1.12 Propositions: A proposition is a declarative sentence (that is, a sentence that declares a fact) that is either true or false, but not both.

All the following declarative sentences are propositions.

Example 1: Washington, D.C., is the capital of the United States of America.

Example 2: Toronto is the capital of Canada.

Example 3: 1 + 1 = 2.

Example 4: 2 + 2 = 3.
Propositions 1 and 3 are true, whereas 2 and 4 are false.

Consider the following sentences.
1. What time is it?
2. Read this carefully.
3. x + 1 = 2.
4. x + y = z.
1 and 2 are not declarative. Only assertive sentences can be propositions.
We use letters to denote propositional variables, that is, variables that represent propositions, just as letters are used to denote numerical variables. The conventional letters used for propositional variables are p, q, r, s

The truth value of a proposition is true, denoted by T, if it is a true proposition, and the truth value of a proposition is false, denoted by F, if it is a false proposition. Propositions that cannot be expressed in terms of simpler propositions are called **atomic propositions**. The area of logic that deals with propositions is called the **propositional calculus** or **propositional logic**. It was first developed systematically by the Greek philosopher Aristotle more than 2300 years ago.

We now turn our attention to methods for producing new propositions from those that we already have. These methods were discussed by the English mathematician George Boole in 1854 in his book The Laws of Thought. Many mathematical statements are constructed by combining one or more propositions. New propositions, called **compound propositions**, are formed from existing propositions using **logical operators**.

Negation

Let p be a proposition. The negation of p, denoted by ¬p (also denoted by $\bar{p}$), is the statement "It is not the case that p."

Other notations you might see are **~p, −p, p′, Np, $\bar{p}$ and !p.**
অর্থাৎ, একে বিপরীতভাবে প্রকাশ করতে চাইলে → **!p, ~p, p′, $\bar{p}$, ¬p.**
আমরা **[¬p]** এভাবে ব্যবহার করবো।

EXAMPLE 3: Find the negation of the proposition "Michael's PC runs Linux."
→ This negation can be more simply expressed as "Michael's PC does not run Linux."

TABLE 1: The Truth Table for the Negation of a Proposition.

p	¬p
T [1]	F [0]
F [0]	T [1]

Conjunction

Let p and q be propositions. The conjunction of p and q, denoted by **p ∧ q**, is the proposition "**p and q.**" The conjunction p ∧ q is true when both p and q are true and is false otherwise.

Ex: It's Thursday and it's raining.

→ এখানে, It's Thursday – True কিন্তু it's raining – false. সেজন্য Output হবে False.

The Truth Table for the Conjunction of Two Propositions.

p	q	p ∧ q
T [1]	T [1]	T [1]
T [1]	F [0]	F [0]
F [0]	T [1]	F [0]
F [0]	F [0]	F [0]

3 parts এর ক্ষেত্রে,

It's thursday and it's sunny and it's 9 am.

এখানে, sunny পর্যন্ত প্রথম ২ অংশ true. এই অংশকে একত্রে নিবো। কিন্তু পরের অংশ false. তাই true and false এর output হবে False.

[উদাহরণগুলো ক্লাস চলাকালীন বুঝানোর সুবিধার্থে ব্যবহার করা হয়েছিল।]

Disjunction

The use of the connective or in a disjunction corresponds to one of the two ways the word or is used in English, namely, as an **inclusive or**. A disjunction is true when at least one of the two propositions is true. That is, **p V q** is true when both p and q are true or when exactly one of p and q is true.

EXAMPLE 6: Translate the statement "Students who have taken calculus or introductory computer science can take this class" in a statement in propositional logic using the propositions p: "A student who has taken calculus can take this class" and q: "A student who has taken introductory computer science can take this class.
Hence, this statement can be expressed as p V q, the inclusive or, or disjunction, of p and q.

Ex: I'm in CSE230 class or in 2nd semester.
→ এখানে, I'm in CSE230 class – True কিন্তু in 2nd semester – false. কেননা class এ সবাই 2nd semester এর না। এখানে Output হবে True.

[উদাহরণগুলো ক্লাস চলাকালীন বুঝানোর সুবিধার্থে ব্যবহার করা হয়েছিল।]

The Truth Table for the Disjunction of Two Propositions.

p	q	p V q
T [1]	T [1]	T [1]
T [1]	F [0]	T [1]
F [0]	T [1]	T [1]
F [0]	F [0]	F [0]

Exclusive OR

Let p and q be propositions. The exclusive or of p and q, denoted by

$p \oplus q$ (or p XOR q), is the proposition that is true when exactly one of p and q is true and is false otherwise.

The Truth Table for the Exclusive Or of Two Propositions.

p	q	$p \oplus q$
T [1]	T [1]	F [0]
T [1]	F [0]	T [1]
F [0]	T [1]	T [1]
F [0]	F [0]	F [0]

EXAMPLE 8: Let p and q be the propositions that state "A student can have a salad with dinner" and "A student can have soup with dinner," respectively. What is $p \oplus q$, the exclusive or of p and q
The exclusive or of p and q is the statement that is true when exactly one of p and q is true. That is, $p \oplus q$ is the statement "A student can have soup or salad, but not both, with dinner." Note that this is often stated as "A student can have soup or a salad with dinner," without explicitly stating that taking both is not permitted.

Ex: You can take Mojo or Clemon.
→ এখানে, যেকোনো ১ টা নিতেই হবে। তখন output True হবে। আবার, ২ টাই নেয়া যাবে না কিংবা কোনোটাই না নিয়ে থাকা যাবে না।

Conditional Statements

Let p and q be propositions. The conditional statement **p → q** is the proposition **"if p, then q"/ "p only if q".** The conditional statement p → q is false when p is true and q is false, and true otherwise. In the conditional statement p → q, p is called the hypothesis (or antecedent or premise) and q is called the conclusion (or consequence).

Other ways to express conditional statements

"if p, then q"	"p implies q"
"if p, q"	"p only if q"
"p is sufficient for q"	"a sufficient condition for q is p"
"q if p"	"q whenever p"
"q when p"	"q is necessary for p"
"a necessary condition for p is q"	"q follows from p"
"q unless $\neg p$"	

"If I am elected, then I will lower taxes."
"If you get 100% on the final, then you will get an A."

Remark: Because some of the different ways to express the implication p implies q can be confusing, we will provide some extra guidance. To remember that "p only if q" expresses the same thing as "if p, then q," note that "p only if q" says that p cannot be true when q is not true. That is, the statement is false if p is true, but q is false. When p is false, q may be either true or false, because the statement says nothing about the truth value of q.

"You can receive an A in the course only if your score on the final is at least 90%.

EXAMPLE 10: Let p be the statement "Maria learns discrete mathematics" and q the statement "Maria will find a good job." Express the statement p → q as a statement in English."

There are many other ways to express this conditional statement in English. Among the most natural of these are “Maria will find a good job when she learns discrete mathematics.” “For Maria to get a good job, it is sufficient for her to learn discrete mathematics.” and “Maria will find a good job unless she does not learn discrete mathematics.”

The Truth Table for the Conditional Statement

p	q	p → q
T [1]	T [1]	T [1]
T [1]	F [0]	F [0]
F [0]	T [1]	T [1]
F [0]	F [0]	T [1]

Hypothesis এ F[0] মানেই output-T[1]. Condition তখনই দেখবো যখন Hypothesis এ কাজ T[1] হচ্ছে। Hypothesis এ T[1] হলে পরে যদি T[1] পাই তাহলে output-T[1]. আর যদি না পাই F[0] হয় তাহলে output-F[0].

Ex:

p – If you get good grades,

q – You will get treat.

Good grades না পেলেও treat দিতে পারে কেননা না পেলে কি করবে সেটা বলেনি।

q unless ¬p

→ You will get treat unless you get bad grades.

ব্যাপারটা এমন যে, যতক্ষণ খারাপ নম্বর পাচ্ছে না, ততক্ষণ treat পাবে।

CONVERSE, CONTRAPOSITIVE, AND INVERSE

We can form some new conditional statements starting with a conditional statement p → q. In particular, there are three related conditional statements that occur so often that they have special names. The proposition q → p is called the converse of p → q. The

contrapositive of p → q is the proposition ¬q → ¬p. The proposition ¬p → ¬ q is called the inverse of p → q. We will see that of these three conditional statements formed from p → q, only the contrapositive always has the same truth value as p → q.

We first show that the contrapositive, ¬q → ¬p, of a conditional statement p → q always has the same truth value as p → q. To see this, note that the contrapositive is false only when ¬p is false and ¬q is true, that is, only when p is true and q is false. We now show that neither the converse, q → p, nor the inverse, ¬p → ¬q, has the same truth value as p → q for all possible truth values of p and q. Note that when p is true and q is false, the original conditional statement is false, but the converse and the inverse are both true.

p → q

- Converse: q → p
- Inverse: ¬p → ¬ q
- Contrapositive: ¬q → ¬p
- Contrapositive = Converse + Inverse

EXAMPLE 12: Find the contrapositive, the converse, and the inverse of the conditional statement Extra Examples "The home team wins whenever it is raining."

Solution: Because "q whenever p" is one of the ways to express the conditional statement p → q, the original statement can be rewritten as "If it is raining, then the home team wins." Consequently, the contrapositive of this conditional statement is "If the home team does not win, then it is not raining." The converse is "If the home team wins, then it is raining." The inverse is "If it is not raining, then the home team does not win." Only the contrapositive is equivalent to the original statement.

p	q	$p \rightarrow q$	$q \rightarrow p$
F [0]	F [0]	T [1]	T [1]
F [0]	T [1]	T [1]	F [0]
T [1]	F [0]	F [0]	T [1]
T [1]	T [1]	T [1]	T [1]

Hypothesis এ F[0] মানেই output-T[1]. Condition তখনই দেখবো যখন Hypothesis এ T[1] হচ্ছে। Hypothesis এ T[1] হলে পরে যদি T[1] পাই তাহলে output-T[1]. আর যদি না পাই F[0] হয় তাহলে output-F[0].

Biconditional Statement

Let p and q be propositions. The biconditional statement **p ↔ q** is the proposition **"p if and only if q."** The biconditional statement p ↔ q is true when p and q have the same truth values, and is false otherwise. Biconditional statements are also called bi-implications.
"p is necessary and sufficient for q" "if p then q, and conversely" "p iff q." "p exactly when q."

p	q	p ↔ q
T [1]	T [1]	T [1]
T [1]	F [0]	F [0]
F [0]	T [1]	F [0]
F [0]	F [0]	T [1]

◆ p ↔ q = (p → q) ∧ (q → p)

অর্থাৎ, ব্যাপারটা এমন যে, তুমি treat দিলে আমিও treat দিবো। তুমি না দিলে আমিও দিবো না। এই ক্ষেত্রেই output True হবে।

Example: Construct the truth table of the compound proposition **(p ∨ ¬q) → (p ∧ q)**

p	q	¬q	p ∨ ¬q	p ∧ q	(p ∨ ¬q) → (p ∧ q)
T [1]	T [1]	F [0]	T [1]	T [1]	T [1]
T [1]	F [0]	T [1]	T [1]	F [0]	F [0]
F [0]	T [1]	F [0]	F [0]	F [0]	T [1]
F [0]	F [0]	T [1]	T [1]	F [0]	F [0]

TABLE 8
Precedence of Logical Operators.

Operator	*Precedence*
$\neg$	1
$\wedge$	2
$\vee$	3
$\rightarrow$	4
$\leftrightarrow$	5

TABLE 9 **Table for the Bit Operators *OR*, *AND*, and *XOR*.**

x	y	$x \vee y$	$x \wedge y$	$x \oplus y$
0	0	0	0	0
0	1	1	0	1
1	0	1	0	1
1	1	1	1	0

◆ A bit string is a sequence of zero or more bits. The length of this string is the number of bits in the string.

EXAMPLE 16 Find the bitwise *OR*, bitwise *AND*, and bitwise *XOR* of the bit strings 01 1011 0110 and 11 0001 1101. (Here, and throughout this book, bit strings will be split into blocks of four bits to make them easier to read.)

Solution: The bitwise *OR*, bitwise *AND*, and bitwise *XOR* of these strings are obtained by taking the *OR*, *AND*, and *XOR* of the corresponding bits, respectively. This gives us

```
01 1011 0110
11 0001 1101
-------------
11 1011 1111   bitwise OR
01 0001 0100   bitwise AND
10 1010 1011   bitwise XOR
```

◀

1.2 Applications of Propositional Logic

EXAMPLE 1 How can this English sentence be translated into a logical expression?

Extra Examples ❯

"You can access the Internet from campus only if you are a computer science major or you are not a freshman."

them. In particular, we let a, c, and f represent "You can access the Internet from campus," "You are a computer science major," and "You are a freshman," respectively. Noting that "only if" is one way a conditional statement can be expressed, this sentence can be represented as

$$a \rightarrow (c \vee \neg f).$$

◀

EXAMPLE 2 How can this English sentence be translated into a logical expression?

"You cannot ride the roller coaster if you are under 4 feet tall unless you are older than 16 years old."

Solution: Let q, r, and s represent "You can ride the roller coaster," "You are under 4 feet tall," and "You are older than 16 years old," respectively. Then the sentence can be translated to

$$(r \wedge \neg s) \rightarrow \neg q.$$

EXAMPLE 7 As a reward for saving his daughter from pirates, the King has given you the opportunity to win a treasure hidden inside one of three trunks. The two trunks that do not hold the treasure are empty. To win, you must select the correct trunk. Trunks 1 and 2 are each inscribed with the message "This trunk is empty," and Trunk 3 is inscribed with the message "The treasure is in Trunk 2." The Queen, who never lies, tells you that only one of these inscriptions is true, while the other two are wrong. Which trunk should you select to win?

Solution: Let p_i be the proposition that the treasure is in Trunk i, for $i = 1, 2, 3$. To translate into propositional logic the Queen's statement that exactly one of the inscriptions is true, we observe that the inscriptions on Trunk 1, Trunk 2, and Trunk 3, are $\neg p_1$, $\neg p_2$, and p_2, respectively. So, her statement can be translated to

$$(\neg p_1 \wedge \neg(\neg p_2) \wedge \neg p_2) \vee (\neg(\neg p_1) \wedge \neg p_2 \wedge \neg p_2) \vee (\neg(\neg p_1) \wedge \neg(\neg p_2) \wedge p_2)).$$

Using the rules for propositional logic, we see that this is equivalent to $(p_1 \wedge \neg p_2) \vee (p_1 \wedge p_2)$. By the distributive law, $(p_1 \wedge \neg p_2) \vee (p_1 \wedge p_2)$ is equivalent to $p_1 \wedge (\neg p_2 \vee p_2)$. But because $\neg p_2 \vee p_2$ must be true, this is then equivalent to $p_1 \wedge \mathbf{T}$, which is in turn equivalent to p_1. So the treasure is in Trunk 1 (that is, p_1 is true), and p_2 and p_3 are false; and the inscription on Trunk 2 is the only true one. (Here, we have used the concept of propositional equivalence, which is discussed in Section 1.3.)

◀

EXAMPLE 8 In [Sm78] Smullyan posed many puzzles about an island that has two kinds of inhabitants, knights, who always tell the truth, and their opposites, knaves, who always lie. You encounter two people A and B. What are A and B if A says "B is a knight" and B says "The two of us are opposite types"?

Extra Examples ❯

1.2.6 Logic Circuits

A **logic circuit** (or **digital circuit**) receives input signals $p_1, p_2, \ldots, p_n$, each a bit [either 0 (off) or 1 (on)], and produces output signals $s_1, s_2, \ldots, s_n$, each a bit. In this section we will restrict our attention to logic circuits with a single output signal; in general, digital circuits may have multiple outputs.

Complicated digital circuits can be constructed from three basic circuits, called **gates**, shown in Figure 1. The **inverter**, or **NOT gate**, takes an input bit p, and produces as output $\neg p$. The **OR gate** takes two input signals p and q, each a bit, and produces as output the signal $p \vee q$. Finally, the **AND gate** takes two input signals p and q, each a bit, and produces as output the signal $p \wedge q$. We use combinations of these three basic gates to build more complicated circuits, such as that shown in Figure 2.

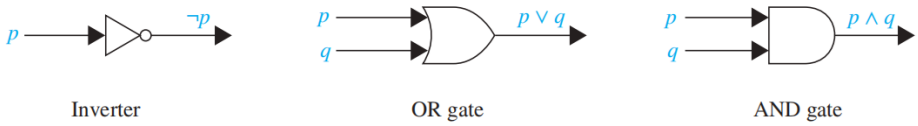


FIGURE 1 **Basic logic gates.**

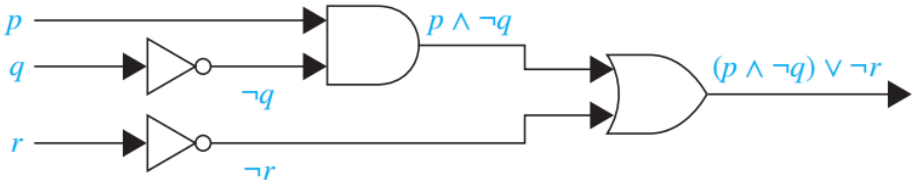


FIGURE 2 **A combinatorial circuit.**

Definition 1 A compound proposition that is always true, no matter what the truth values of the propositional variables that occur in it, is called a *tautology*. A compound proposition that is always false is called a *contradiction*. A compound proposition that is neither a tautology nor a contradiction is called a *contingency*.

EXAMPLE 1 We can construct examples of tautologies and contradictions using just one propositional variable. Consider the truth tables of $p \vee \neg p$ and $p \wedge \neg p$, shown in Table 1. Because $p \vee \neg p$ is always true, it is a tautology. Because $p \wedge \neg p$ is always false, it is a contradiction. ◀

TABLE 1 Examples of a Tautology and a Contradiction.

p	$\neg p$	$p \vee \neg p$	$p \wedge \neg p$
T	F	T	F
F	T	T	F

Definition 2 The compound propositions p and q are called *logically equivalent* if $p \leftrightarrow q$ is a tautology. The notation $p \equiv q$ denotes that p and q are logically equivalent.

TABLE 2 De Morgan's Laws.

$$\neg(p \wedge q) \equiv \neg p \vee \neg q$$

$$\neg(p \vee q) \equiv \neg p \wedge \neg q$$

EXAMPLE 2 Show that $\neg(p \vee q)$ and $\neg p \wedge \neg q$ are logically equivalent.

TABLE 3 Truth Tables for $\neg(p \vee q)$ and $\neg p \wedge \neg q$.

p	q	$p \vee q$	$\neg(p \vee q)$	$\neg p$	$\neg q$	$\neg p \wedge \neg q$
T	T	T	F	F	F	F
T	F	T	F	F	T	F
F	T	T	F	T	F	F
F	F	F	T	T	T	T

EXAMPLE 3 Show that $p \to q$ and $\neg p \vee q$ are logically equivalent. (This is known as the **conditional-disjunction equivalence**.)

TABLE 4 Truth Tables for $\neg p \vee q$ and $p \to q$.

p	q	$\neg p$	$\neg p \vee q$	$p \to q$
T	T	F	T	T
T	F	F	F	F
F	T	T	T	T
F	F	T	T	T

TABLE 6 Logical Equivalences.

Equivalence	*Name*
$p \wedge \mathbf{T} \equiv p$ $p \vee \mathbf{F} \equiv p$	Identity laws
$p \vee \mathbf{T} \equiv \mathbf{T}$ $p \wedge \mathbf{F} \equiv \mathbf{F}$	Domination laws
$p \vee p \equiv p$ $p \wedge p \equiv p$	Idempotent laws
$\neg(\neg p) \equiv p$	Double negation law

$p \vee q \equiv q \vee p$ $p \wedge q \equiv q \wedge p$	Commutative laws
$(p \vee q) \vee r \equiv p \vee (q \vee r)$ $(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$	Associative laws
$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$ $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$	Distributive laws
$\neg(p \wedge q) \equiv \neg p \vee \neg q$ $\neg(p \vee q) \equiv \neg p \wedge \neg q$	De Morgan's laws
$p \vee (p \wedge q) \equiv p$ $p \wedge (p \vee q) \equiv p$	Absorption laws
$p \vee \neg p \equiv \mathbf{T}$ $p \wedge \neg p \equiv \mathbf{F}$	Negation laws

The statement "x is greater than 3" has two parts. The first part, the variable x, is the subject of the statement. The second part—the **predicate**, "is greater than 3"—refers to a property that the subject of the statement can have. We can denote the statement "x is greater than 3" by $P(x)$, where P denotes the predicate "is greater than 3" and x is the variable. The statement $P(x)$ is also said to be the value of the **propositional function** P at x. Once a value has been assigned to the variable x, the statement $P(x)$ becomes a proposition and has a truth value. Consider Examples 1 and 2.

EXAMPLE 1 Let $P(x)$ denote the statement "$x > 3$." What are the truth values of $P(4)$ and $P(2)$?

Solution: We obtain the statement $P(4)$ by setting $x = 4$ in the statement "$x > 3$." Hence, $P(4)$, which is the statement "$4 > 3$," is true. However, $P(2)$, which is the statement "$2 > 3$," is false. ◀

EXAMPLE 2 Let $A(x)$ denote the statement "Computer x is under attack by an intruder." Suppose that of the computers on campus, only CS2 and MATH1 are currently under attack by intruders. What are truth values of $A(\text{CS1})$, $A(\text{CS2})$, and $A(\text{MATH1})$?

Solution: We obtain the statement $A(\text{CS1})$ by setting $x = \text{CS1}$ in the statement "Computer x is under attack by an intruder." Because CS1 is not on the list of computers currently under attack, we conclude that $A(\text{CS1})$ is false. Similarly, because CS2 and MATH1 are on the list of computers under attack, we know that $A(\text{CS2})$ and $A(\text{MATH1})$ are true. ◀

Ex: P(x) = x is good at Maths.

এখানে, ক্লাস এ সবাই সমানভাবে Math এ ভালো না। যারা Math এ ভালো তাদের ক্ষেত্রে True. আর যারা ভালো না তাদের ক্ষেত্রে False.

EXAMPLE 3 Let $Q(x, y)$ denote the statement "$x = y + 3$." What are the truth values of the propositions $Q(1, 2)$ and $Q(3, 0)$?

Extra Examples

Solution: To obtain $Q(1, 2)$, set $x = 1$ and $y = 2$ in the statement $Q(x, y)$. Hence, $Q(1, 2)$ is the statement "$1 = 2 + 3$," which is false. The statement $Q(3, 0)$ is the proposition "$3 = 0 + 3$," which is true. ◀

1.4.3 Quantifiers

Definition 1 The *universal quantification* of $P(x)$ is the statement

"$P(x)$ for all values of x in the domain."

The notation $\forall x P(x)$ denotes the universal quantification of $P(x)$. Here $\forall$ is called the **universal quantifier**. We read $\forall x P(x)$ as "for all $xP(x)$" or "for every $xP(x)$." An element for which $P(x)$ is false is called a **counterexample** to $\forall x P(x)$.

TABLE 1 Quantifiers.

Statement	*When True?*	*When False?*
$\forall x P(x)$	$P(x)$ is true for every x.	There is an x for which $P(x)$ is false.
$\exists x P(x)$	There is an x for which $P(x)$ is true.	$P(x)$ is false for every x.

Ex: P(x) = All the x students of the class are good at math.

∀xP(x). অর্থাৎ, ক্লাসের সবাই (১০০%) Math এ ভালো হলেই True হবে। ১ জনও যদি Math এ খারাপ হয় তাহলে False.

Definition 2 The *existential quantification* of $P(x)$ is the proposition

"There exists an element x in the domain such that $P(x)$."

We use the notation $\exists x P(x)$ for the existential quantification of $P(x)$. Here $\exists$ is called the *existential quantifier*.

Ex: P(x) = There are x students in the class and they are good at math.

∃xP(x). অর্থাৎ, ক্লাসের ১ জন Math এ ভালো হলেই True হবে। সবাই (১০০%) যদি Math এ খারাপ হয় শুধুমাত্র তখনই False.

1.4.9 Negating Quantified Expressions

We will often want to consider the negation of a quantified expression. For instance, consider the negation of the statement

"Every student in your class has taken a course in calculus."

This statement is a universal quantification, namely,

$\forall x P(x)$,

We will often want to consider the negation of a quantified expression. For instance, consider the negation of the statement "Every student in your class has taken a course in calculus."

This statement is a universal quantification, namely,

∀xP(x)

where P(x) is the statement “x has taken a course in calculus” and the domain consists of the students in your class. The negation of this statement is “It is not the case that every student in your class has taken a course in calculus.” This is equivalent to “There is a student in your class who has not taken a course in calculus.” And this is simply the existential quantification of the negation of the original propositional function, namely,

∃x ¬P(x).

This example illustrates the following logical equivalence:

¬∀xP(x) ≡ ∃x ¬P(x).

একটু ভেঙ্গে দেখলে,

P(x) = x has taken a course in calculus.

¬P(x) = It is not the case that every student in your class has taken a course in calculus.

∀xP(x) = Every student in your class has taken a course in calculus.

¬∀xP(x) = Not every student in your class has taken a course in calculus.

∃x¬P(x) = There is a student in your class who has not taken a course in calculus.

অর্থাৎ, **¬∀xP(x)** দিয়ে বুঝানো হচ্ছে যে ক্লাস এ সবাই calculus course নেয়নি।

একইভাবে, **∃x¬P(x)** দিয়ে বুঝানো হচ্ছে যে, এই কথাটা কারো কারো জন্য সত্য যে, তারা calculus course নেয়নি। পারতপক্ষে, ২ টা একই কথা।

¬∀xP(x) ≡ ∃x ¬P(x).

Suppose we wish to negate an existential quantification. For instance, consider the proposition “There is a student in this class who has taken a course in calculus.” This is the existential quantification

∃xQ(x)

where Q(x) is the statement “x has taken a course in calculus.” The negation of this statement is the proposition “It is not the case that there is a student in this class who has taken a course in calculus.” This is equivalent to “Every student in this class has not taken calculus,” which is just the universal quantification of the negation of the original propositional function, or, phrased in the language of quantifiers,

∀x ¬Q(x).

This example illustrates the equivalence

¬∃xQ(x) ≡ ∀x ¬Q(x).

TABLE 2 **De Morgan’s Laws for Quantifiers.**

Negation	*Equivalent Statement*	*When Is Negation True?*	*When False?*
$\neg\exists x P(x)$	$\forall x \neg P(x)$	For every x, $P(x)$ is false.	There is an x for which $P(x)$ is true.
$\neg\forall x P(x)$	$\exists x \neg P(x)$	There is an x for which $P(x)$ is false.	$P(x)$ is true for every x.

EXAMPLE 24 Express the statements "Some student in this class has visited Mexico" and "Every student in this class has visited either Canada or Mexico" using predicates and quantifiers.

→ We introduce M(x), which is the statement "x has visited Mexico." If the domain for x consists of the students in this class, we can translate this first statement as ∃xM(x).

(Note that we are assuming the inclusive, rather than the exclusive, or here.) We let C(x) be "x has visited Canada." Following our earlier reasoning, we see that if the domain for x consists of the students in this class, this second statement can be expressed as ∀x(C(x) ∨ M(x)).

একটু অন্যভাবে দেখি,

S(x) = "x is a student of this class."

M(x) = "x has visited Mexico."/

M(x,y) = "x has visited y."

C(x) = "x has visited Canada."

(i) "Some student in this class has visited Mexico."

Sol: ∃x(S(x) ∧ M(x))

(ii) "Every student in this class has visited either Canada or Mexico."

Sol: ∀x(S(x) → (C(x) ∨ M(x)))/

∀x(S(x) → (M(x, "Mexico") ∨ M(x, "Canada")))

Nested Quantifiers

Note that everything within the scope of a quantifier can be thought of as a propositional function. For example,

∀x∃y(x + y = 0)

is the same thing as ∀xQ(x), where Q(x) is ∃yP (x, y), where P (x, y) is x + y = 0.

বোঝাচ্ছে যে, সকল x এর জন্য কোনো না কোনো y আছে যাতে x + y = 0 হয়।

অর্থাৎ, প্রত্যেক x এর ই (-x) আছে যা x – x = 0 হয়।

EXAMPLE 1 Assume that the domain for the variables x and y consists of all real numbers. The statement

$$\forall x \forall y(x + y = y + x)$$

Extra Examples says that $x + y = y + x$ for all real numbers x and y. This is the commutative law for addition of real numbers. Likewise, the statement

$$\forall x \exists y(x + y = 0)$$

says that for every real number x there is a real number y such that $x + y = 0$. This states that every real number has an additive inverse. Similarly, the statement

$$\forall x \forall y \forall z(x + (y + z) = (x + y) + z)$$

is the associative law for addition of real numbers. ◀

EXAMPLE 2 Translate into English the statement

$$\forall x \forall y((x > 0) \wedge (y < 0) \rightarrow (xy < 0)),$$

where the domain for both variables consists of all real numbers.

EXAMPLE 4 Let $Q(x, y)$ denote "$x + y = 0$." What are the truth values of the quantifications $\exists y \forall x Q(x, y)$ and $\forall x \exists y Q(x, y)$, where the domain for all variables consists of all real numbers?

EXAMPLE 5 Let $Q(x, y, z)$ be the statement "$x + y = z$." What are the truth values of the statements $\forall x \forall y \exists z Q(x, y, z)$ and $\exists z \forall x \forall y Q(x, y, z)$, where the domain of all variables consists of all real numbers?

Translate the statement "The sum of two positive integers is always positive" into a logical expression.

$$\forall x \forall y((x > 0) \wedge (y > 0) \rightarrow (x + y > 0)),$$

where the domain for both variables consists of all integers.

$$\forall x \forall y(x + y > 0),$$

where the domain for both variables consists of all positive integers.

EXAMPLE 7 Translate the statement "Every real number except zero has a multiplicative inverse." (A **multiplicative inverse** of a real number x is a real number y such that $xy = 1$.)

Solution: We first rewrite this as "For every real number x except zero, x has a multiplicative inverse." We can rewrite this as "For every real number x, if $x \neq 0$, then there exists a real number y such that $xy = 1$." This can be rewritten as

$$\forall x((x \neq 0) \rightarrow \exists y(xy = 1)).$$ ◀

EXAMPLE 8 (*Requires calculus*) Use quantifiers to express the definition of the limit of a real-valued function $f(x)$ of a real variable x at a point a in its domain.

Solution: Recall that the definition of the statement

$$\lim_{x \to a} f(x) = L$$

is: For every real number $\epsilon > 0$ there exists a real number $\delta > 0$ such that $|f(x) - L| < \epsilon$ whenever $0 < |x - a| < \delta$. This definition of a limit can be phrased in terms of quantifiers by

$$\forall \epsilon \exists \delta \forall x (0 < |x - a| < \delta \rightarrow |f(x) - L| < \epsilon),$$

where the domain for the variables δ and ϵ consists of all positive real numbers and for x consists of all real numbers.

This definition can also be expressed as

$$\forall \epsilon > 0 \; \exists \delta > 0 \; \forall x (0 < |x - a| < \delta \rightarrow |f(x) - L| < \epsilon)$$

when the domain for the variables ϵ and δ consists of all real numbers, rather than just the positive real numbers. [Here, restricted quantifiers have been used. Recall that $\forall x > 0 \, P(x)$ means that for all x with $x > 0$, $P(x)$ is true.] ◀

1.5.5 Translating from Nested Quantifiers into English

EXAMPLE 9: Translate the statement

∀x(C(x) ∨ ∃y(C(y) ∧ F(x, y))) into English, where C(x) is "x has a computer," F(x, y) is "x and y are friends," and the domain for both x and y consists of all students in your school.

Solution: The statement says that for every student x in your school, x has a computer or there is a student y such that y has a computer and x and y are friends. In other words, every student in your school has a computer or has a friend who has a computer.

◆ Translate the statement

∀x(I(x) ∨ ∃y(I(y) ∧ F(x, y))) into English, where I(x) is "x has a iPhone," F(x, y) is "x and y are friends," and the domain for both x and y consists of all students.

Solution:

∀x(I(x) ∨ ∃y(I(y) ∧ F(x, y))) বোঝাচ্ছে, For All x বা পৃথিবীর প্রত্যেকটা মানুষের জন্য হয় x এর নিজস্ব iPhone আছে **[I(x)]** অথবা অন্য একজন y মানুষ আছে [**∃y**] যে y এর iPhone আছে [**I(y)**] এবং সে x এর বন্ধু **[F(x, y)]**।

1.5.6 Translating English Sentences into Logical Expressions

EXAMPLE 11 Express the statement "If a person is female and is a parent, then this person is someone's mother" as a logical expression involving predicates, quantifiers with a domain consisting of all people, and logical connectives.

Solution: The statement "If a person is female and is a parent, then this person is someone's mother" can be expressed as "For every person x, if person x is female and person x is a parent, then there exists a person y such that person x is the mother of person y." We introduce the propositional functions $F(x)$ to represent "x is female," $P(x)$ to represent "x is a parent," and $M(x, y)$ to represent "x is the mother of y." The original statement can be represented as

$$\forall x((F(x) \wedge P(x)) \rightarrow \exists y M(x, y)).$$

EXAMPLE 12 Express the statement "Everyone has exactly one best friend" as a logical expression involving predicates, quantifiers with a domain consisting of all people, and logical connectives.

Solution: The statement "Everyone has exactly one best friend" can be expressed as "For every person x, person x has exactly one best friend." Introducing the universal quantifier, we see that this statement is the same as "$\forall x$(person x has exactly one best friend)," where the domain consists of all people.

To say that x has exactly one best friend means that there is a person y who is the best friend of x, and furthermore, that for every person z, if person z is not person y, then z is not the best friend of x. When we introduce the predicate $B(x, y)$ to be the statement "y is the best friend of x," the statement that x has exactly one best friend can be represented as

$$\exists y(B(x, y) \wedge \forall z((z \neq y) \rightarrow \neg B(x, z))).$$

Consequently, our original statement can be expressed as

$$\forall x \exists y(B(x, y) \wedge \forall z((z \neq y) \rightarrow \neg B(x, z))).$$

EXAMPLE 13 Use quantifiers to express the statement "There is a woman who has taken a flight on every airline in the world."

Solution: Let $P(w, f)$ be "w has taken f" and $Q(f, a)$ be "f is a flight on a." We can express the statement as

$$\exists w \forall a \exists f(P(w, f) \wedge Q(f, a)),$$

1.5.7 Negating Nested Quantifiers

EXAMPLE 14 Express the negation of the statement $\forall x \exists y(xy = 1)$ so that no negation precedes a quantifier.

Extra Examples

Solution: By successively applying De Morgan's laws for quantifiers in Table 2 of Section 1.4, we can move the negation in $\neg\forall x \exists y(xy = 1)$ inside all the quantifiers. We find that $\neg\forall x \exists y(xy = 1)$ is equivalent to $\exists x \neg\exists y(xy = 1)$, which is equivalent to $\exists x \forall y \neg(xy = 1)$. Because $\neg(xy = 1)$ can be expressed more simply as $xy \neq 1$, we conclude that our negated statement can be expressed as $\exists x \forall y(xy \neq 1)$. ◀

EXAMPLE 15 Use quantifiers to express the statement that "There does not exist a woman who has taken a flight on every airline in the world."

Solution: This statement is the negation of the statement "There is a woman who has taken a flight on every airline in the world" from Example 13. By Example 13, our statement can be expressed as $\neg\exists w \forall a \exists f(P(w, f) \wedge Q(f, a))$, where $P(w, f)$ is "w has taken f" and $Q(f, a)$ is "f is a flight on a." By successively applying De Morgan's laws for quantifiers in Table 2 of Section 1.4 to move the negation inside successive quantifiers and by applying De Morgan's law for negating a conjunction in the last step, we find that our statement is equivalent to each of this sequence of statements:

$$\begin{aligned}\forall w \neg\forall a \exists f(P(w, f) \wedge Q(f, a)) &\equiv \forall w \exists a \neg\exists f(P(w, f) \wedge Q(f, a)) \\ &\equiv \forall w \exists a \forall f \neg(P(w, f) \wedge Q(f, a)) \\ &\equiv \forall w \exists a \forall f(\neg P(w, f) \vee \neg Q(f, a)).\end{aligned}$$

This last statement states "For every woman there is an airline such that for all flights, this woman has not taken that flight or that flight is not on this airline." ◀

EXAMPLE 16 (*Requires calculus*) Use quantifiers and predicates to express the fact that $\lim_{x \to a} f(x)$ does not exist where $f(x)$ is a real-valued function of a real variable x and a belongs to the domain of f

Solution: To say that $\lim_{x \to a} f(x)$ does not exist means that for all real numbers L, $\lim_{x \to a} f(x) \neq L$. By using Example 8, the statement $\lim_{x \to a} f(x) \neq L$ can be expressed as

$$\neg\forall \epsilon > 0\ \exists \delta > 0\ \forall x(0 < |x - a| < \delta \rightarrow |f(x) - L| < \epsilon).$$

Successively applying the rules for negating quantified expressions, we construct this sequence of equivalent statements:

$$\begin{aligned}&\neg\forall \epsilon > 0\ \exists \delta > 0\ \forall x(0 < |x - a| < \delta \rightarrow |f(x) - L| < \epsilon) \\ &\quad \equiv \exists \epsilon > 0\ \neg\exists \delta > 0\ \forall x(0 < |x - a| < \delta \rightarrow |f(x) - L| < \epsilon) \\ &\quad \equiv \exists \epsilon > 0\ \forall \delta > 0\ \neg\forall x(0 < |x - a| < \delta \rightarrow |f(x) - L| < \epsilon) \\ &\quad \equiv \exists \epsilon > 0\ \forall \delta > 0\ \exists x\ \neg(0 < |x - a| < \delta \rightarrow |f(x) - L| < \epsilon) \\ &\quad \equiv \exists \epsilon > 0\ \forall \delta > 0\ \exists x(0 < |x - a| < \delta \wedge |f(x) - L| \geq \epsilon).\end{aligned}$$

In the last step we used the equivalence $\neg(p \rightarrow q) \equiv p \wedge \neg q$,

Proof Techniques

Introduction to Proofs: Terminologies

- Formally, a **theorem** is a statement that can be shown to be true. In mathematical writing, the term theorem is usually reserved for a statement that is considered at least somewhat important. Less important theorems sometimes are called **propositions**. (Theorems can also be referred to as **facts** or **results**.)
- A theorem may be the universal quantification of a conditional statement with one or more premises and a conclusion. However, it may be some other type of logical statement, as the examples later in this chapter will show.
- We demonstrate that a theorem is true with a **proof**. A proof is a valid argument that establishes the truth of a theorem.
- The statements used in a proof can include
  - **axioms** (or **postulates**), which are statements we assume to be true (for example, the axioms for the real numbers, given in Appendix 1, and the axioms of plane geometry),
  - the premises, if any, of the theorem,
  - and previously proven theorems.
- Axioms may be stated using primitive terms that do not require definition, but all other terms used in theorems and their proofs must be defined.
- Rules of inference, together with definitions of terms, are used to draw conclusions from other assertions, tying together the steps of a proof.
- In practice, the final step of a proof is usually just the conclusion of the theorem. However, for clarity, we will often

recap the statement of the theorem as the final step of a proof.

- A less important theorem that is helpful in the proof of other results is called a **lemma** (plural *lemmas* or *lemmata*).
- Complicated proofs are usually easier to understand when they are proved using a series of lemmas, where each lemma is proved individually.
- A **corollary** is a theorem that can be established directly from a theorem that has been proved.
- A **conjecture** is a statement that is being proposed to be a true statement, usually on the basis of some partial evidence, a heuristic argument, or the intuition of an expert.
- When a proof of a conjecture is found, the conjecture becomes a theorem.
- Many times conjectures are shown to be false, so they are not theorems.

Direct Proof: Procedure

- A direct proof of a conditional statement $p \rightarrow q$ is constructed when
  - the first step is the assumption that p is true;
  - subsequent steps are constructed using rules of inference,
  - with the final step showing that q must also be true.
- A direct proof shows that a conditional statement $p \rightarrow q$ is true by showing that if p is true, then q must also be true, so that the combination p true and q false never occurs.

Direct Proof: examples

For this example, we need to define even and odd integers.

- The integer n is even if there exists an integer k such that $n = 2k$, and n is odd if there exists an integer k such that $n = 2k + 1$.

- Note that every integer is either even or odd, and no integer is both even and odd.
- Two integers have the same parity when both are even or both are odd; they have opposite parity when one is even and the other is odd.

◆ Now, give a direct proof of the theorem "If n is an odd integer, then n^2 is odd."

Sol: $n = 2k + 1$

$\Rightarrow n^2 = (2k + 1)^2$

$\Rightarrow n^2 = 4k^2 + 4k + 1$

$\Rightarrow n^2 = 2k(2k + 2) + 1$

◆ Give a direct proof that if m and n are both perfect squares, then nm is also a perfect square. (An integer a is a perfect square if there is an integer b such that $a = b^2$.)

Solution: To produce a direct proof of this theorem, we assume that the hypothesis of this conditional statement is true, namely, we assume that m and n are both perfect squares. By the definition of a perfect square, it follows that there are integers s and t such that $m = s^2$ and $n = t^2$. The goal of the proof is to show that mn must also be a perfect square when m and n are; looking ahead we see how we can show this by substituting s^2 for m and t^2 for n into mn. This tells us that $mn = s^2t^2$. Hence, $mn = s^2t^2 = (ss)(tt) = (st)(st) = (st)^2$, using commutativity and associativity of multiplication. By the definition of perfect square, it follows that mn is also a perfect square, because it is the square of st, which is an integer. We have proved that if m and n are both perfect squares, then mn is also a perfect square.

Proof by Contraposition: procedure

- Proofs by contraposition make use of the fact that the conditional statement p → q is equivalent to its contrapositive, ¬q → ¬p.
- This means that the conditional statement p → q can be proved by showing that its contrapositive, ¬q → ¬p, is true.
- In a proof by contraposition of p → q, we take ¬q as a premise, and using axioms, definitions, and previously proven theorems, together with rules of inference, we show that ¬p must follow.

◆ Prove that if n is an integer and $3n + 2$ is odd, then n is odd.

Solution: We first attempt a direct proof. To construct a direct proof, we first assume that $3n + 2$ is an odd integer. From the definition of an odd integer, we know that $3n + 2 = 2k + 1$ for some integer k. Can we use this fact to show that n is odd? We see that $3n + 1 = 2k$, but there does not seem to be any direct way to conclude that n is odd. Because our attempt at a direct proof failed, we next try a proof by contraposition.

Because our attempt at a direct proof failed, we next try a proof by contraposition.

The first step in a proof by contraposition is to assume that the conclusion of the conditional statement "If $3n + 2$ is odd, then n is odd" is false; namely, assume that n is even. Then, by the definition of an even integer, $n = 2k$ for some integer k. Substituting $2k$ for n, we find that $3n + 2 = 3(2k) + 2 = 6k + 2 = 2(3k + 1)$. This tells us that $3n + 2$ is even (because it is a multiple of 2), and therefore not odd. This is the negation of the premise of the theorem. Because the

negation of the conclusion of the conditional statement implies that the hypothesis is false, the original conditional statement is true. Our proof by contraposition succeeded; we have proved the theorem "If $3n + 2$ is odd, then n is odd."

এখানে, p → q এর contrapositive, ¬q → ¬p হওয়ায় conclusion কে false অর্থাৎ, n কে even ধরে নিবো। তাহলে,

n = 2k
3n + 2
= 3(2k) + 2
= 6k + 2
= 2(3k + 1) (Proved)

◆ Prove that if $n = ab$, where a and b are positive integers, then $a \leq \sqrt{n}$ or $b \leq \sqrt{n}$.
Because there is no obvious way of showing that $a \leq \sqrt{n}$ or $b \leq \sqrt{n}$ directly from the equation $n = ab$, where a and b are positive integers, we attempt a proof by contraposition. The first step in a proof by contraposition is to assume that the conclusion of the conditional statement "If $n = ab$, where a and b are positive integers, then $a \leq \sqrt{n}$ or $b \leq \sqrt{n}$" is false. That is, we assume that the statement $(a \leq \sqrt{n}) \vee (b \leq \sqrt{n})$ is false. Using the meaning of disjunction together with De Morgan's law, we see that this implies that both $a \leq \sqrt{n}$ and $b \leq \sqrt{n}$ are false. This implies that $a > \sqrt{n}$ and $b > \sqrt{n}$.

We can multiply these inequalities together (using the fact that if $0 < s < t$ and $0 < u < v$, then $su < tv$) to obtain $ab > \sqrt{n} \cdot \sqrt{n} = n$. This shows that $ab \neq n$, which contradicts the statement $n = ab$. Because the negation of the conclusion of the conditional statement implies that the hypothesis is false, the original conditional statement is true. Our

proof by contraposition succeeded; we have proved that if $n = ab$, where a and b are positive integers, then $a \leq \sqrt{n}$ or $b \leq \sqrt{n}$.

Proof strategy: examples

For this example, we need to define rational and irrational numbers.

- The real number r is rational if there exist integers p and q with $q \neq 0$ such that $r = p/q$. A real number that is not rational is called irrational.

Now prove that the sum of two rational numbers is rational. (Note that if we include the implicit quantifiers here, the theorem we want to prove is "For every real number r and every real number s, if r and s are rational numbers, then $r + s$ is rational.)

Solution: We first attempt a direct proof. To begin, suppose that r and s are rational numbers. From the definition of a rational number, it follows that there are integers p and q, with $q \neq 0$, such that $r = p/q$, and integers t and u, with $u \neq 0$, such that $s = t/u$.

Can we use this information to show that $r + s$ is rational? That is, can we find integers v and w such that $r + s = v/w$ and $w \neq 0$? With the goal of finding these integers v and w, we add $r = p/q$ and $s = t/u$, using qu as the common denominator. We find that $r + s = p/q + t/u = (pu + qt)/qu$.

Because $q \neq 0$ and $u \neq 0$, it follows that $qu \neq 0$. Consequently, we have expressed $r + s$ as the ratio of two integers, $v = pu + qt$ and $w = qu$, where $w \neq 0$. This means that $r + s$ is rational. We have proved that the sum of two rational numbers is rational; our attempt to find a direct proof succeeded.

Proof by Contradiction: procedure

Suppose we want to prove that a statement p is true. Furthermore, suppose that we can find a contradiction q such that ¬p → q is true. Because *q* is false, but ¬p → q is true, we can conclude that ¬p is false, which means that *p* is true. How can we find a contradiction q that might help us prove that p is true in this way? [অর্থাৎ, p যদি false হয় তাহলে কোনো না কোনো ঝামেলা বাঁধবেই। সেই ঝামেলার নাম দিলাম q. এখানে, q কিন্তু আমাদের প্রমাণ করার অংশ না। Contradiction সবসময় false.]

Because the statement r ∧ ¬r is a contradiction whenever *r* is a proposition, we can prove that *p* is true if we can show that ¬p → (r ∧ ¬r) is true for some proposition *r*. Proofs of this type are called proofs by contradiction.

Proof by Contradiction: examples

Show that at least four of any 22 days must fall on the same day of the week.

Solution: Let p be the proposition "At least four of 22 chosen days fall on the same day of the week." Suppose that $\neg p$ is true. This means that at most three of the 22 days fall on the same day of the week. Because there are seven days of the week, this implies that at most 21 days could have been chosen, as for each of the days of the week, at most three of the chosen days could fall on that day. This contradicts the premise that we have 22 days under consideration. That is, if r is the statement that 22 days are chosen, then we have shown that $\neg p \rightarrow (r \wedge \neg r)$. Consequently, we know that p is true. We have proved that at least four of 22 chosen days fall on the same day of the week.

◆ Prove that $\sqrt{2}$ is irrational by giving a proof by contradiction.

Solution: Let p be the proposition "$\sqrt{2}$ is irrational." To start a proof by contradiction, we suppose that $\neg p$ is true. Note that $\neg p$ is the statement "It is not the case that $\sqrt{2}$ is irrational," which says that $\sqrt{2}$ is rational. We will show that assuming that $\neg p$ is true leads to a contradiction.

If $\sqrt{2}$ is rational, there exist integers a and b with $\sqrt{2} = a/b$, where $b \neq 0$ and a and b have no common factors (so that the fraction a/b is in lowest terms). (Here, we are using the fact that every rational number can be written in lowest terms.) Because $\sqrt{2} = a/b$, when both sides of this equation are squared, it follows that $2 = a^2/b^2$. Hence, $2b^2 = a^2$.

By the definition of an even integer it follows that a^2 is even. We next use the fact that if a^2 is even, a must also be even (which follows by

Exercise 18).

Furthermore, because a is even, by the definition of an even integer, $a = 2c$ for some integer c. Thus, $2b^2 = 4c^2$. Dividing both sides of this equation by 2 gives $b^2 = 2c^2$. By the definition of even, this means that b^2 is even. Again using the fact that if the square of an integer is even, then the integer itself must be even, we conclude that b must be even as well.

We have now shown that the assumption of $\neg p$ leads to the equation $\sqrt{2} = a/b$, where a and b have no common factors, but both a and b are even, that is, 2 divides both a and b. Note that the statement that $\sqrt{2} = a/b$, where a and b have no common factors, means, in particular, that 2 does not divide both a and b. Because our assumption of $\neg p$ leads to the contradiction that 2 divides both a and b and 2 does not divide both a and b, $\neg p$ must be false. That is, the statement p, "$\sqrt{2}$ is irrational," is true. We have proved that $\sqrt{2}$ is irrational.

সহজে বললে, $\sqrt{2}$ একটি অমূলদ সংখ্যা, সেটা প্রমাণ করতে হবে।

$$\sqrt{2} = \frac{a}{b}$$

$$\Rightarrow 2 = \frac{a^2}{b^2}$$

$$\Rightarrow 2b^2 = a^2$$

Proof by Counterexample: procedure

- To show that a statement of the form $\forall x P(x)$ is false, we need only find a counterexample, that is, an example x for which P(x) is false.
- When presented with a statement of the form $\forall x P(x)$, which we believe to be false or which has resisted all proof attempts, we look for a counterexample.

Proof by Counterexample: examples

Show that the statement "Every positive integer is the sum of the squares of two integers" is false.

Solution: To show that this statement is false, we look for a counterexample, which is a particular integer that is not the sum of the squares of two integers. It does not take long to find a counterexample, because 3 cannot be written as the sum of the squares of two integers. To show this is the case, note that the only perfect squares not exceeding 3 are $0^2 = 0$ and $1^2 = 1$. Furthermore, there is no way to get 3 as the sum of two terms each of which is 0 or 1. Consequently, we have shown that "Every positive integer is the sum of the squares of two integers" is false.

Mistakes in Proofs

Mistakes in Proofs: examples

What is wrong with this famous supposed "proof" that 1 = 2?

Proof: We use these steps, where a and b are two equal positive integers.

Step	Reason
1. $a = b$	Given
2. $a^2 = ab$	Multiply both sides of (1) by a
3. $a^2 - b^2 = ab - b^2$	Subtract b^2 from both sides of (2)
4. $(a - b)(a + b) = b(a - b)$	Factor both sides of (3)
5. $a + b = b$	Divide both sides of (4) by $a - b$
6. $2b = b$	Replace a by b in (5) because $a = b$ and simplify
7. $2 = 1$	Divide both sides of (6) by b

Mistakes in Proofs: incorrect conclusion

- A common error of reasoning is to draw incorrect conclusions from examples.
- No matter how many separate examples are considered, a theorem is not proved by considering examples unless every possible case is covered.
- The problem of proving a theorem is analogous to showing that a computer program always produces the output desired.
- No matter how many input values are tested, unless all input values are tested, we cannot conclude that the program always produces the correct output.

Mistakes in Proofs: incorrect conclusion examples

Is it true that every positive integer is the sum of 18 fourth powers of integers?

Solution: To determine whether a positive integer n can be written as the sum of 18 fourth powers of integers, we might begin by examining whether n is the sum of 18 fourth powers of integers for the smallest positive integers. Because the fourth powers of integers are 0, 1, 16, 81, … , if we can select 18 terms from these numbers that add up to n, then n is the sum of 18 fourth powers.

We can show that all positive integers up to 78 can be written as the sum of 18 fourth powers. (The details are left to the reader.) However, if we decided this was enough checking, we would come to the wrong conclusion. It is not true that every positive integer is the sum of 18 fourth powers because 79 is not the sum of 18 fourth powers (as the reader can verify).

Existence Proofs

- Many theorems are assertions that objects of a particular type

exist. A theorem of this type is a proposition of the form $\exists x P(x)$, where P is a predicate.

- A proof of a proposition of the form $\exists x P(x)$ is called an **existence proof**. There are several ways to prove a theorem of this type.
- Sometimes an existence proof of $\exists x P(x)$ can be given by finding an element a, called a witness, such that $P(a)$ is true. This type of existence proof is called constructive.
- It is also possible to give an existence proof that is non-constructive; that is, we do not find an element a such that $P(a)$ is true, but rather prove that $\exists x P(x)$ is true in some other way.
- One common method of giving a non-constructive existence proof is to use proof by contradiction and show that the negation of the existential quantification implies a contradiction.

Constructive and non-constructive Proofs: examples

Show that there is a positive integer that can be written as the sum of cubes of positive integers in two different ways.

Solution: After considerable computation (such as a computer search) we can find that $1729 = 10^3 + 9^3 = 12^3 + 1^3$.

Because we have displayed a positive integer that can be written as the sum of cubes in two different ways, we are done.

Show that there exist irrational numbers x and y such that x^y is rational.

Solution: We know that $\sqrt{2}$ is irrational. Consider the number $(\sqrt{2})^{\sqrt{2}}$. If it is rational, we have two irrational numbers x and y with x^y rational, namely, $x = \sqrt{2}$ and $y = \sqrt{2}$. On the other hand if $(\sqrt{2})^{\sqrt{2}}$ is irrational, then we can let $x = (\sqrt{2})^{\sqrt{2}}$ and $y = \sqrt{2}$ so that $x^y = ((\sqrt{2})^{\sqrt{2}})^{\sqrt{2}} =$

$(\sqrt{2})^{(\sqrt{2}\cdot\sqrt{2})} = (\sqrt{2})^2 = 2$.

This proof is an example of a non-constructive existence proof because we have not found irrational numbers x and y such that x^y is rational. Rather, we have shown that either the pair $x = \sqrt{2}$, $y = \sqrt{2}$ or the pair $x = (\sqrt{2})^{\sqrt{2}}$, $y = \sqrt{2}$ have the desired property, but we do not know which of these two pairs works!

Open problems

- **FERMAT'S LAST THEOREM**: The equation $x^n + y^n = z^n$ has no solutions in integers x, y, and z with $xyz \neq 0$ whenever n is an integer with $n > 2$.
- **The 3x + 1 Conjecture**: Let T be the transformation that sends an even integer x to $x/2$ and an odd integer x to $3x + 1$. A famous conjecture, sometimes known as the $3x + 1$ conjecture, states that for all positive integers x, when we repeatedly apply the transformation T, we will eventually reach the integer 1. For example, starting with x = 13, we find $T(13) = 3 \cdot 13 + 1 = 40$, $T(40) = 40/2 = 20$, $T(20) = 20/2 = 10$, $T(10) = 10/2 = 5$, $T(5) = 3 \cdot 5 + 1 = 16$, $T(16) = 8$, $T(8) = 4$, $T(4) = 2$, and $T(2) = 1$. The 3x + 1 conjecture has been verified using computers for all integers x up to $5.48 \cdot 10^{18}$.
- **Twin Prime Conjecture**: There are infinitely many pairs of twin primes.

Rules of Inference

- By an argument, we mean a sequence of statements that end with a conclusion.
- By valid, we mean that the conclusion, or final statement of the argument, must follow from the truth of the preceding statements, or premises, of the argument. That is, an argument

is valid if and only if it is impossible for all the premises to be true and the conclusion to be false.

- To deduce new statements from statements we already have, we use rules of inference which are templates for constructing valid arguments. Rules of inference are our basic tools for establishing the truth of statements of Inference.
- After we illustrate how rules of inference are used to produce valid arguments, we will describe some common forms of incorrect reasoning, called fallacies, which lead to invalid arguments.
- An **argument** in propositional logic is a sequence of propositions.
- All but the final proposition in the argument are called **premises** and the final proposition is called the **conclusion**.
- An argument is **valid** if the truth of all its premises implies that the conclusion is true.
- An **argument form** in propositional logic is a sequence of compound propositions involving propositional variables.
- An argument form is **valid** if no matter which particular propositions are substituted for the propositional variables in its premises, the conclusion is true if the premises are all true.

Determine whether the argument given here is valid and determine whether its conclusion must be true because of the validity of the argument.

"If $\sqrt{2} > 3/2$, then $(\sqrt{2})^2 > (3/2)^2$.

We know that $\sqrt{2} > 3/2$. Consequently, $(\sqrt{2})^2 = 2 > (3/2)^2 = 9/4$."

Solution: Let p be the proposition "$\sqrt{2} > 3/2$" and q the proposition "$2 > (3/2)^2$." The premises of the argument are $p \rightarrow q$ and p, and q is its conclusion. This argument is valid because it is constructed by using

modus ponens, a valid argument form. However, one of its premises, $2 > 3/2$, is false. Consequently, we cannot conclude that the conclusion is true. Furthermore, note that the conclusion of this argument is false, because $2 < 9/4$.

Rules of Inference: Modus Ponens

p

$p \rightarrow q$

———

$\therefore q$

where $\therefore$ is the symbol that denotes "therefore."

Rules of Inference: Modus Tollens

$\neg q$

$p \rightarrow q$

———

$\therefore \neg p$

where $\therefore$ is the symbol that denotes "therefore."

Rules of Inference: Addition

p

———

$\therefore p \vee q$

where $\therefore$ is the symbol that denotes "therefore."

Rules of Inference: Simplification

$p \wedge q$

———

$\therefore p$

where $\therefore$ is the symbol that denotes "therefore."

Fallacies: fallacy of affirming the conclusion

Is the following argument valid?

If you do every problem in this book, then you will learn discrete mathematics. You learned discrete mathematics.

Therefore, you did every problem in this book.

Solution: Let p be the proposition "You did every problem in this book." Let q be the proposition "You learned discrete mathematics."

Then this argument is of the form: if $(p \rightarrow q)$ and q, then p.

This is an example of an incorrect argument using the fallacy of affirming the conclusion. Indeed, it is possible for you to learn discrete mathematics in some way other than by doing every problem in this book. (You may learn discrete mathematics by reading, listening to lectures, doing some, but not all, the problems in this book, and so on.)

CHAPTER 2: SETS

Definition: A set is an unordered collection of distinct objects, called elements or members of the set. A set is said to contain its elements. We write a ∈ A to denote that a is an element of the set A. The notation a ∉ A denotes that a is not an element of the set A.
There are several ways to describe a set. One way is to list all the members of a set, when this is possible. We use a notation where all members of the set are listed between braces. For example, the notation {a, b, c, d} represents the set with the four elements a, b, c, and d. This way of describing a set is known as the roster method.

Examples

EXAMPLE 1: The set V of all vowels in the English alphabet can be written as **V = {a, e, i, o, u}**.

EXAMPLE 2: The set O of odd positive integers less than 10 can be expressed by **O = {1, 3, 5, 7, 9}**.

EXAMPLE 3: The set L of Nobel Laureates in chemistry in Bangladesh, **L = { }**, known as empty set often denoted by ϕ.

Elements of a set may even be inconceivably unrelated like {a, 2, Sabuj, Rajshahi}

Set Builder Notation

Another way to describe a set is to use set builder notation. We characterize all those Extra elements in the set by stating the property or properties they must have to be members. General form

of this notation is **{ x | x has property P}** and is read "the set of all x such that x has property P." For instance, the set O of all even positive integers less than 10 can be written as **O = { x | x is an even positive integer less than 10 }**, that can also be enumerated as **{0, 2, 4, 6, 8}**. However, if it were 100 instead of 10, it would have been difficult to do so.

Some of the accepted notations are as follows:

- **N** = {1, 2, 3, …}, the set of all **Natural numbers**
- **Z** = {… ,−2,−1, 0, 1, 2, …}, the set of all **Integers**
- $\mathbf{Z^+}$ = {1, 2, 3, …}, the set of all **Positive Integers**
- **Q** = {p/q | p ∈ **Z,** q ∈ **Z,** and q ≠ 0}, the set of all **Rational Numbers**
- **R**, the set of all **Real Numbers**
- $\mathbf{R^+}$, the set of all **Positive Real Numbers**
- **C**, the set of all **Complex Numbers.**

If a and b are real numbers with a ≤ b, we denote these intervals by:

- ❑ [a, b] = { x | a ≤ x ≤ b }
- ❑ [a, b) = { x | a ≤ x < b }
- ❑ (a, b] = { x | a < x ≤ b }
- ❑ (a, b) = { x | a < x < b }

Note that **[a, b] is called the closed interval** from a to b and **(a, b) is called the open interval** from a to b.

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$[a, b] = \{ x \mid a \leq x \leq b \}$
$[a, b) = \{ x \mid a \leq x < b \}$
$(a, b] = \{ x \mid a < x \leq b \}$
$(a, b) = \{ x \mid a < x < b \}$

$[3, 5]$



$(a, b]$



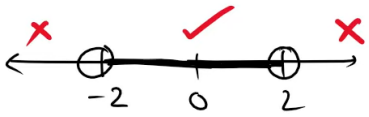
$[3, 5)$



$x^2 < 4$

$x^2 - 4 < 0$

$(x+2)(x-2) < 0$



$S = (-2, 2)$

Set Equality

Definition : Two sets are equal if and only if they have the same elements. Therefore, if A and B are sets, then A and B are equal if and only if
$\forall x (x \in A \leftrightarrow x \in B)$. We write A = B, if A and B are equal sets.

EXAMPLE: The sets {1, 3, 7} and {3, 7, 1} are equal, because they have the same elements. Note that the order in which the elements of a set are listed does not matter. Note also that it does not matter if an element of a set is listed more than once, so {2, 3, 3, 3, 5, 5, 5} is the same as the set {2, 3, 5} because they have the same elements.

Empty Set

There is a special set that has no elements. This set is called the empty set, or null set, and is denoted by ϕ or **{ }**.

A set with one element is called a **singleton set.**

A common error is to confuse the empty {ϕ} has one more element than ϕ. Set ϕ with the set {ϕ} , which is a **singleton set**.

The single element of the set {ϕ} is the **empty set** itself!

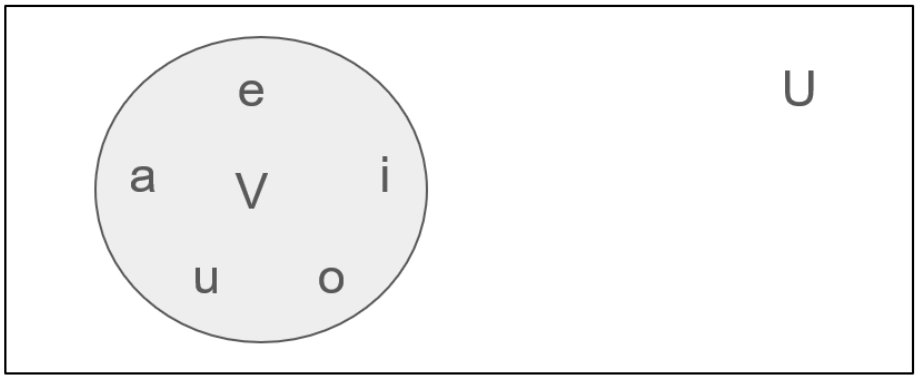
ϕ, {ϕ} and {{ϕ}} are all different sets.

◆ **{ϕ} / {{ }}** মানে ফাঁকা সেট নয়। সেট এর ভিতর ১ টা element আছে এবং সেটা ফাঁকা সেট।

Venn Diagram

Sets can be represented graphically using Venn diagrams, named after the English mathematician John Venn, who introduced their use in 1881. In Venn diagrams the universal set U, which contains all the objects under consideration, is represented by a rectangle.

EXAMPLE: Draw a Venn diagram that represents V, the set of vowels in the English alphabet.



Subsets

Definition : The set A is a subset of B, and B is a superset of A, if and only if every element of A is also an element of B. We use the notation **A ⊆ B** to indicate that **A is a subset of the set B.** If, instead, we want to stress that **B is a superset of A**, we use the equivalent notation **B ⊇ A.** (So, **A ⊆ B** and **B ⊇ A** are equivalent statements.)

We see that **A ⊆ B** if and only if the quantification **∀ x(x ∈ A → x ∈ B)**

Showing that A is a Subset of B To show that **A ⊆ B**, show that if x belongs to A then x also belongs to B.
অর্থাৎ, A এর সবগুলো উপাদান যদি B এর মধ্যে থাকে তখন ই A, B এর subset হবে।

A = {1, 3, 5}
B = {1, 3, 5, 7, 9}
এখানে, **A ⊆ B**

Showing that A is Not a Subset of B To show that **A ⊈ B**, find a single **x ∈ A** such that **x ∉ B**
অর্থাৎ, A এর সবগুলো উপাদান যদি B এর মধ্যে না থাকে তখন A, B এর subset হবে না।

A = {1, 3, 5, 6}
B = {1, 3, 5, 7, 9}
এখানে, **A ⊈ B**

Theorems

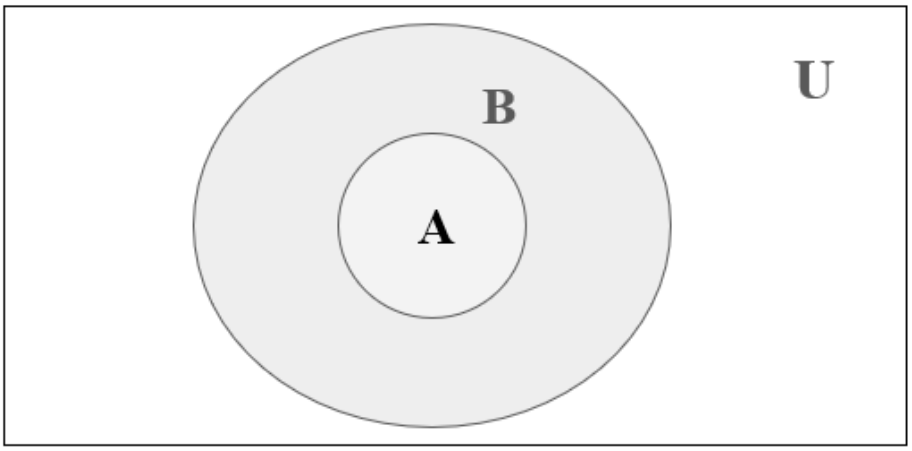
THEOREM 1: For every set S, (i) ∅ ⊆ S and (ii) S ⊆ S.

Proof: We will prove **(i)** and leave the proof of **(ii)** as an exercise.

Let **S** be a set. To show that **∅ ⊆ S**, we must show that **∀x(x ∈ ∅ → x ∈ S)** is true. Because the empty set contains no elements, it follows that **x ∈ ∅** is always false. It follows that the conditional statement **x ∈ ∅ → x ∈ S** is always true, because its hypothesis is always false and a conditional statement with a false hypothesis is true.
Therefore, **∀x(x ∈ ∅ → x ∈ S)** is true. This completes the proof of **(i)**.
[Note that this is an example of a vacuous proof.]

Proper Subset

When we wish to emphasize that a set A is a subset of a set B but that **A ≠ B**, we write **A ⊂ B** and say that A is a proper subset of B. For **A ⊂ B** to be true, it must be the case that **A ⊆ B** and there must exist an element x of B that is not an element of A. That is, A is a proper subset of B if and only if
∀ X(x ∈ A → x ∈ B) ∧ ∃ x(x ∈ B ∧ x ∉ A)



The Size of a Set

Sets may have other sets as members. For instance, we have the sets **A = {∅, {a}, {b}, {a, b}}** and **B = {x | x is a subset of the set {a, b}}.** Note that these two sets are equal, that is, **A = B.** Also note that **{a} ∈ A**, but **a ∉ A.**

Definition : Let S be a set. If there are exactly n distinct elements in S where n is a nonnegative integer, we say that S is a finite set and that n is the cardinality of S. The cardinality of S is denoted by **|S|.**

EXAMPLE 10: Let A be the set of odd positive integers less than 10. Then **|A| = 5.**

EXAMPLE 11: Let S be the set of letters in the English alphabet. Then **|S| = 26.**

EXAMPLE 12: Because the null set has no elements, it follows that **|∅| = 0.**

Infinite sets and power sets

Definition : A set is said to be infinite if it is not finite.
EXAMPLE: The set of positive integers is infinite.

Definition : Given a set S, the power set of S is the set of all subsets of the set S. The power set of S is denoted by (S).

EXAMPLE: What is the power set of the set {0, 1, 2}?
Solution: The power set ({0, 1, 2}) is the set of all subsets of {0, 1, 2}. Hence, P ({0, 1, 2}) = {∅, {0}, {1}, {2}, {0, 1}, {0, 2}, {1, 2}, {0, 1, 2}}.

Note that the empty set and the set itself are members of this set of subsets.

Power set of empty set

EXAMPLE : What is the power set of the empty set? What is the power set of the set **{ Ø }**?

Solution: The empty set has exactly one subset, namely, itself. Consequently,
P(Ø) = {Ø}

The set **{Ø}** has exactly two subsets, namely, **Ø** and the set **{Ø}** itself. Therefore,
P({Ø}) = {Ø, {Ø} }

If a set has n elements, then its **power set has 2^n elements.**

Cartesian Products

Definition: The ordered n-tuple $(a_1, a_2, \dots, a_n)$ is the ordered collection that has a_1 as its first element, a_2 as its second element, … , and a_n as its nth element.

We say that two ordered n-tuples are equal if and only if each corresponding pair of their elements is equal. In other words, $(a_1, a_2, \dots, a_n) = (b_1, b_2, \dots, b_n)$ if and only if $a_i = b_i$ **, for** $i = 1, 2, \dots, n$.

Definition: Let A and B be sets. The Cartesian product of A and B, denoted by $A \times B$, is the set of all ordered pairs (a, b), where $a \in A$ and $b \in B$.

Hence, $A \times B = \{(a, b) \mid a \in A \wedge b \in B\}$.

Examples

EXAMPLE: What is the Cartesian product of A = {1, 2} and B = {a, b, c}?

Solution:

The Cartesian product $A \times B$ is: $A \times B = \{(1, a), (1, b), (1, c), (2, a), (2, b), (2, c)\}$.

Note that the Cartesian products $A \times B$ and $B \times A$ are not equal unless $A = \emptyset$ or $B = \emptyset$ (so that $A \times B = \emptyset$) or $A = B$

Definition: The Cartesian product of the sets $A_1, A_2, \cdots, A_n$, denoted by $A_1 \times A_2 \times \cdots \times A_n$, is the set of ordered n-tuples $(a_1, a_2, \dots, a_n)$, where a_i belongs to A_i for $i = 1, 2, \dots, n$. In other words, $A_1 \times A_2 \times \cdots \times A_n = \{(a_1, a_2, \dots, a_n) \mid a_i \in A_i \text{ for } i = 1, 2, \dots, n\}$.

What is the Cartesian product **A × B × C**, where **A = {0, 1}, B = {1, 2}, and C = {0, 1, 2}?**
Solution: The Cartesian product **A × B × C** consists of all ordered triples (a, b, c), where **a ∈ A, b ∈ B**, and **c ∈ C.**

এখানে, A, B এর ১ম উপাদানগুলোকে [0, 1] ধ্রুবক বা constant হিসেবে ধরে নিবো তারপর একে একে C এর উপাদানগুলোকে আলাদা করবো।

Hence, **A × B × C = {(0, 1, 0), (0, 1, 1), (0, 1, 2), (0, 2, 0), (0, 2, 1), (0, 2, 2), (1, 1, 0), (1, 1, 1), (1, 1, 2), (1, 2, 0), (1, 2, 1), (1, 2, 2)}.**

We use the notation A^2 to denote A × A, the Cartesian product of the set A with itself. Similarly, A^3 = A × A × A, A^4 = A × A × A × A, and so on. More generally, $A^n = \{(a_1, a_2, \dots, a_n) \mid a_i \in A \text{ for } i = 1, 2, \dots, n\}$.

EXAMPLE: Suppose that A = {1, 2}. It follows that:
A^2 = {(1, 1), (1, 2), (2, 1), (2, 2)} and
A^3 = {(1, 1, 1), (1, 1, 2), (1, 2, 1), (1, 2, 2), (2, 1, 1), (2, 1, 2), (2, 2, 1), (2, 2, 2)}.

EXAMPLE: What are the ordered pairs in the less than or equal to relation, which contains **(a, b) if a ≤ b, on the set {0, 1, 2, 3}?**

Solution: The ordered pair (a, b) belongs to R if and only if both a and b belong to
{0, 1, 2, 3} and a ≤ b.

Consequently, **R = {(0, 0), (0, 1), (0, 2), (0, 3), (1, 1), (1, 2), (1, 3), (2,**

2), (2, 3), (3, 3)}.

EXAMPLE: What do the statement $\forall x \in R\ (x^2 \geq 0)$ mean?

Solution: The statement $\forall x \in R\ (x^2 \geq 0)$ states that for every real number x, $x^2 \geq 0$. This statement can be expressed as "The square of every real number is nonnegative."

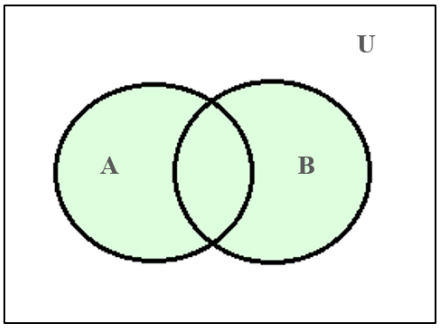
This is a true statement.

2.2 Set Operations

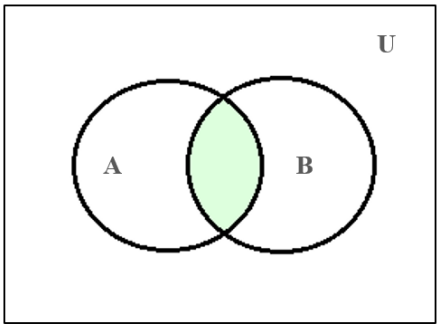
Definition 1: Let A and B be sets. The union of the sets A and B, denoted by A ∪ B, is the set that contains those elements that are either in A or in B, or in both. An element x belongs to the union of the sets A and B if and only if x belongs to A or x belongs to B.
This tells us that **$A \cup B = \{x \mid x \in A \vee x \in B\}$.**

EXAMPLE 1: The union of the sets {1, 3, 5} and {1, 2, 3} is the set {1, 2, 3, 5}; that is,
$\{1, 3, 5\} \cup \{1, 2, 3\} = \{1, 2, 3, 5\}$

EXAMPLE 3: The intersection of the sets {1, 3, 5} and {1, 2, 3} is the set {1, 3}; that is,
$\{1, 3, 5\} \cap \{1, 2, 3\} = \{1, 3\}$.



A ∪ B is shaded.



A ∩ B is shaded.

Disjoint Set

Definition 3: Two sets are called disjoint if their intersection is the empty set.

EXAMPLE 5:

Let A = {1, 3, 5, 7, 9} and B = {2, 4, 6, 8, 10}. Because A ∩ B = ∅, A and B are disjoint.

Principle of inclusion–exclusion

We are often interested in finding the cardinality of a union of two finite sets A and B. Note that **|A| + |B|** counts each element that is in A but not in B or in B but not in A exactly once, and Be careful not to overcount! each element that is in both A and B exactly twice. Thus, if the number of elements that are in both A and B is subtracted from |A| + |B|, elements in A ∩ B will be counted only once.

Hence, **|A ∪ B| = |A| + |B| − |A ∩ B|.**
[এখানে, **|A|** দ্বারা A set এর size বোঝাচ্ছে।]

The generalization of this result to unions of an arbitrary number of sets is called the principle of **inclusion–exclusion.** The principle of inclusion–exclusion is an important technique used in enumeration. We will discuss this principle and other counting techniques in detail in Chapters 6 and 8.

Difference of two sets

Definition 4: Let A and B be sets. The difference of A and B, denoted by A − B, is the set containing those elements that are in A but not in B. The difference of A and B is also called the complement of B with respect to A.

Remark: The difference of sets A and B is sometimes denoted by **A\B**. An element x belongs to the difference of A and B if and only if $x \in A$ and $x \notin B$. This tells us that

$A - B = \{x \mid x \in A \wedge x \notin B\}$.

EXAMPLE 6: The difference of {1, 3, 5} and {1, 2, 3} is the set {5}; that is, {1, 3, 5} − {1, 2, 3} = {5}. This is different from the difference of {1, 2, 3} and {1, 3, 5}, which is the set {2}.

Ex:

A = {1, 2, 5, 6}, B = {2, 6, 10, 12}

A - B = {1, 5}

B – A = {10, 12}

Extra elements, Result এ আসবে না। কারণ যেটা থেকে বাদ দিচ্ছি সেটায় সেগুলো নেই।

Complement of a set

Definition 5: Let U be the universal set. The complement of the set A, denoted by A^c, is the complement of A with respect to U.

Therefore, the complement of the set A is U − A.

Remark: The definition of the complement of A depends on a particular universal set U. This definition makes sense for any superset U of A. If we want to identify the universal set U, we would

write “the complement of A with respect to the set U.” An element belongs to A^c if and only if **$x \notin A$.**
This tells us that A^c **$= \{x \in U \mid x \notin A\}$**.

Ex:

$U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$

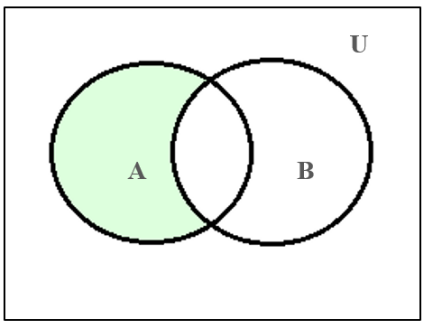
$A = \{1, 2, 5, 6\}$

$A^c = U - A$

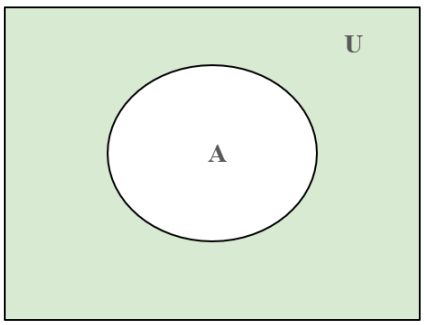
$= \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\} - \{1, 2, 5, 6\}$

$= \{3, 4, 7, 8, 9, 10\}$

$(A^c)^c = U - A^c$



Venn diagram for the difference of A and B



Venn diagram for the complement of the set A.

2.2.2 Set Identities

Identity	Name
$A \cap U = A$ $A \cup \emptyset = A$	Identity laws
$A \cup U = U$ $A \cap \emptyset = \emptyset$	Domination laws
$A \cup A = A$ $A \cap A = A$	Idempotent laws
$(A^c)^c = A$	Complementation law
$A \cup B = B \cup A$ $A \cap B = B \cap A$	Commutative laws
$A \cup (B \cup C) = (A \cup B) \cup C$	Associative laws

$A \cap (B \cap C) = (A \cap B) \cap C$	
$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$	Distributive laws
$(A \cap B)^c = A^c \cup B^c$ $(A \cup B)^c = A^c \cap B^c$	De Morgan's laws
$A \cup (A \cap B) = A$ $A \cap (A \cup B) = A$	Absorption laws
$A \cup \bar{A} = U$ $A \cap \bar{A} = \emptyset$	Complement laws

Proof of DeMorgan's theorem

EXAMPLE 11 Use set builder notation and logical equivalences to establish the first De Morgan law $(A \cap B)^c = A^c \cup B^c$

Solution: We can prove this identity with the following steps.

$(A \cap B)^c = \{x \mid x \notin A \cap B\}$ by definition of complement

$= \{x \mid \neg (x \in (A \cap B))\}$ by definition of does not belong symbol

$= \{x \mid \neg (x \in A \wedge x \in B)\}$ by definition of intersection

$= \{x \mid \neg (x \in A) \vee \neg(x \in B)\}$ by the first De Morgan law for logical equivalences

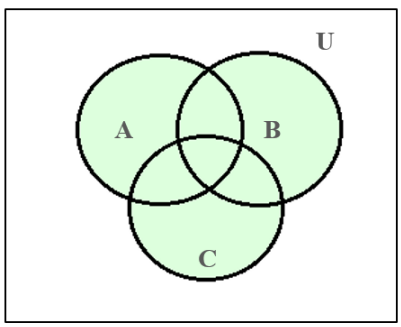
$= \{x \mid x \notin A \vee x \notin B\}$ by definition of does not belong symbol

$= \{x \mid x \in A^c \vee x \in B^c\}$ by definition of complement

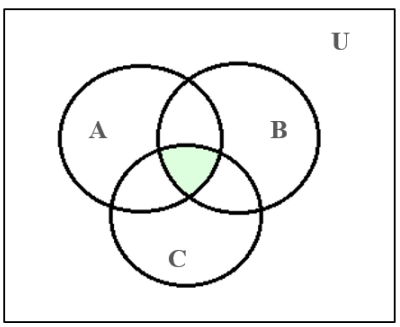
$= \{x \mid x \in A^c \cup B^c\}$ by definition of union

$= A^c \cup B^c$ by meaning of set builder notation

Union and Intersection of several sets



(a) A U B U C is shaded.



(b) A ∩ B ∩ C is shaded.

Ex 13: Use a membership table to show that

$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$.

A	B	C	B U C	A ∩ (B U C)	(A ∩ B)	(A ∩ C)	(A ∩ B) U (A ∩ C)
0	0	0	0	0	0	0	0
0	0	1	1	0	0	0	0
0	1	0	1	0	0	0	0
0	1	1	1	0	0	0	0
1	0	0	0	0	0	0	0
1	0	1	1	1	0	1	1
1	1	0	1	1	1	0	1
1	1	1	1	1	1	1	1

এখানে, **U** মানে Disjunction ($\vee$).

$\cap$ মানে Conjunction ($\wedge$).

EXAMPLE 14 : Let A, B, and C be sets. Let A, B, and C be sets.
Show that $(A \cup (B \cap C))^c = (C^c \cup B^c) \cap A^c$.

Solution: We have $(A \cup (B \cap C))^c$

$= A^c \cap (B \cap C)^c$ by the first De Morgan law

$= A^c \cap (B^c \cup C^c)$ by the second De Morgan law

$= (B^c \cup C^c) \cap A^c$ by the commutative law for intersections

$= (C^c \cup B^c) \cap A^c$ by the commutative law for unions.

Union and intersection of a collection of sets

Definition 6 The union of a collection of sets is the set that contains those elements that are members of at least one set in the collection.

We use the notation $A_1 \cup A_2 \cup \cdots \cup A_n$

$= \bigcup_{i=1}^{n} A_i$ to denote the union of the sets $A_1, A_2, \ldots, A_n$

Definition 7 The intersection of a collection of sets is the set that contains those elements that are members of all the sets in the collection.

We use the notation $A_1 \cap A_2 \cap \cdots \cap A_n = \cap_{i=1}^{n} A_i$

Function

Function Definition

Let ***A*** and ***B*** be nonempty sets. A function ***f*** from ***A*** to ***B*** is an assignment of exactly one element of ***B*** to each element of ***A***. We write ***f(a) = b*** if ***b*** is the unique element of ***B*** assigned by the function ***f*** to the element ***a*** of ***A***. If ***f*** is a function from ***A*** to ***B***, we write ***f : A → B***.

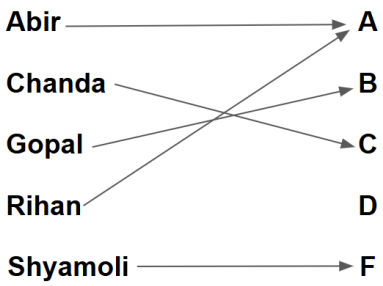


Fig : Assignment of grades in a discrete mathematics class

অর্থাৎ, Different input এর জন্য same output থাকতে পারে কিন্তু ১ টা input এর জন্য একাধিক output থাকতে পারে না, তবেই সেটা একটি ফাংশন।

Domain and Range

If ***f*** is a function from ***A*** to ***B***, we say that ***A*** is the ***domain*** of ***f*** and ***B*** is the ***codomain*** of ***f***. If ***f(a) = b***, we say that ***b*** is the ***image*** of ***a*** and ***a*** is a ***preimage*** of ***b***. The ***range***, or ***image***, of ***f*** is the ***set of all images*** of elements of ***A***. Also, if ***f*** is a function from ***A*** to ***B***, we say that ***f maps A to B***.

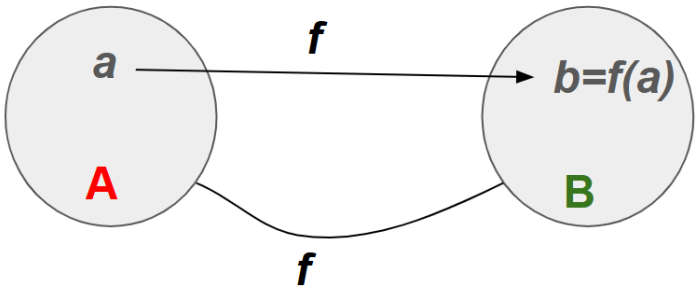
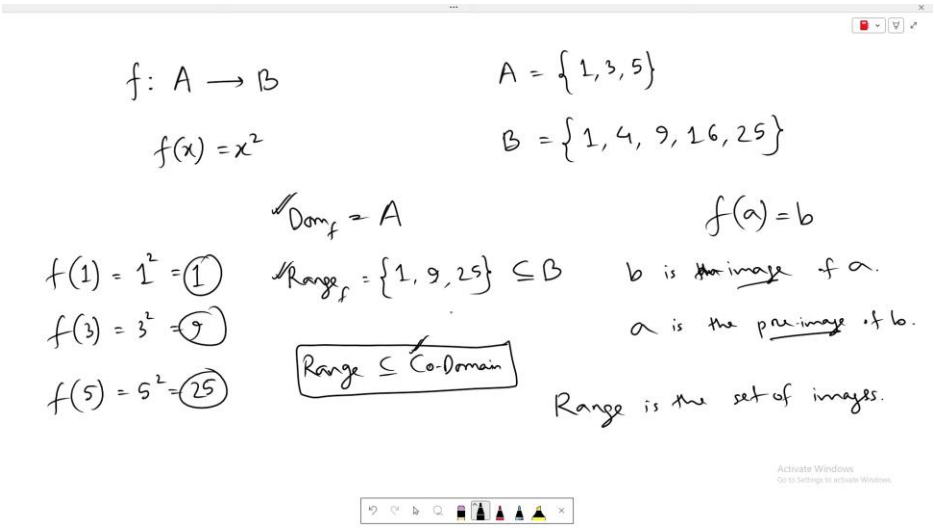


Fig : The function *f* maps *A* to *B*

- Co-Domain = set of possible outputs.
- Range is a subset of Co-Domain.



Examples

EXAMPLE: Let **R** be the relation with ordered pairs (Abdul, 22), (Bidyut, 24), (Chanda, 21), (Dayem, 22), (Emon, 24), and (Fahim, 22). Here each pair consists of a graduate student and this student's age. Specify a function determined by this relation.

SOLUTION: If ***f*** is a function specified by **R**, then ***f(***Abdul***)*** = 22, ***f(***Bidyut***)*** = 24 and so on. For domain we take the set **{Abdul, Bidyut, Chanda, Dayem, Emon, Fahim}**. We also need to specify the codomain, which needs to contain all possible ages of the students. Because it is highly likely that all students are less than 100 years old, we can take the **set of positive integers less than 100** as the codomain.

Definition

Let f_1 and f_2 be functions from ***A*** to ***R***. Then f_1+f_2 and f_1f_2 are also functions from ***A*** to ***R*** defined for all x ∈ A by

- $(f_1+f_2)(x) = f_1(x) + f_2(x)$
- $(f_1f_2)(x) = f_1(x)f_2(x)$

EXAMPLE: Let f_1 and f_2 be functions from ***R*** to ***R*** such that $f_1(x) = x^2$ and $f_2(x) = x - x^2$. What are the functions $f_1 + f_2$ and $f_1 f_2$?

SOLUTION: From the definition of the sum and product of functions, it follows that

➔ $(f_1+ f_2)(x) = f_1(x) + f_2(x) = x^2 + (x - x^2) = x$

➔ $(f_1f_2)(x) = x^2 (x - x^2) = x^3 - x^4$.

Let ***f*** be a function from ***A*** to ***B*** and let ***S*** be a subset of ***A***. The ***image of S*** under the function ***f*** is the ***subset of B*** that consists of the images

of the elements of ***S***. We denote the image of ***S*** by ***f(S)***, so
f(S) = {t | ∃s ∈ S (t = f(s))}.
We also use the shorthand ***{f(s) / s ∈S}*** to denote this set.

EXAMPLE: Let **A** = {a, b, c, d, e} and **B** = {1, 2, 3, 4} with *f(a)* = 2, *f(b)* = 1, *f(c)* = 4, *f(d)* = 1, and *f(e)* = 1. The image of the subset **S** = {b, c, d} is the set *f(S)* = {1, 4}.

One-to-one Functions [Injection]

A function ***f*** is said to be ***one-to-one***, or an ***injection***, if and only if ***f(a) = f(b)*** implies that ***a = b*** for all ***a*** and ***b*** in the domain of ***f***.
A function is said to be injective if it is one-to-one.

Contrapositive of the previous implication: A function ***f*** is ***one-to-one*** if and only if ***f(a) ≠ f(b)*** whenever ***a ≠ b***.

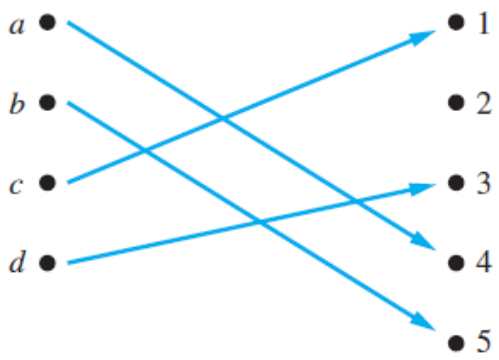
Expressing the previous implication using quantifiers: ∀a∀b (f(a) = f(b) → a = b) or equivalently **∀a∀b(a ≠ b → f(a) ≠ f(b))**, where the universe of discourse is the domain of the function.

◆ ১ টা input এর জন্য ১ টা output.

Examples

EXAMPLE: Determine whether the function ***f*** from **{a, b, c, d}** to **{1, 2, 3, 4, 5}** with ***f(a) = 4, f(b) = 5, f(c) = 1,*** and ***f(d) = 3*** is one-to-one.

SOLUTION: The function ***f*** is one-to-one because ***f*** takes on different values at the four elements of its domain.



EXAMPLE: Determine whether the function $f(x) = x^2$ from the set of integers to the set of integers is one-to-one.

SOLUTION: The function $f(x) = x^2$ is not one-to-one because, for instance, f(1) = f(−1) = 1, but 1 ≠ −1.

EXAMPLE: Determine whether the function $f(x) = x + 1$ from the set of real numbers to itself is one-to-one.

SOLUTION: Suppose that x and y are real numbers with $f(x) = f(y)$, so, f(x) = f(y)

⟶ x + 1 = y + 1

⟶ x = y.

Hence, $f(x) = x + 1$ is a one-to-one function from R to R.

Definition

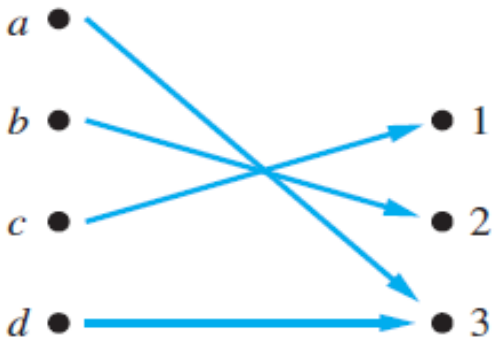
A function f whose domain and codomain are subsets of the set of real numbers is called ***increasing*** if $f(x) \leq f(y)$, and ***strictly increasing*** if $f(x) < f(y)$, whenever $x < y$ and x and y are in the domain of f. Similarly, f is called ***decreasing*** if $f(x) \geq f(y)$, and ***strictly decreasing*** if $f(x) > f(y)$, whenever $x < y$ and x and y are in the domain of f. (The word strictly in this definition indicates a strict inequality.)

EXAMPLE: The function $f(x) = x^2$ from R^+ to R^+ is strictly increasing. To see this, suppose that x and y are positive real numbers with $x < y$. Multiplying both sides of this inequality by x gives $x^2 < xy$. Similarly, multiplying both sides by y gives $xy < y^2$. Hence, $f(x) = x^2 < xy < y^2 =$

f(y). However, the function ***f(x) = x²*** from ***R*** to the set of ***nonnegative real numbers*** is not strictly increasing because ***−1 < 0***, but ***f(−1) = (−1)² = 1*** is not less than ***f(0) = 0² = 0***.

Onto Functions [Surjection]

A function ***f*** from ***A*** to ***B*** is called ***onto***, or a ***surjection***, if and only if for every element ***b ∈ B*** there is an element ***a ∈ A*** with ***f(a) = b***. A function ***f*** is called surjective if it is onto.



- Input ছাড়া কোনো output থাকতে পারবে না।
- Range = Co-Domain

EXAMPLE: Is the function ***f(x) = x+1*** from the set of integers to the set of integers onto?

SOLUTION: This function is onto, because for every integer *y* there is an integer *x* such that *f(x) = y*. To see this, note that *f(x) = y* if and only if *x+1 = y*, which holds if and only if *x = y−1*. (Note that *y−1* is also an integer, and so, is in the domain of *f*).

EXAMPLE: Is the function ***f(x) = x²*** from the set of integers to the set of integers onto?

SOLUTION: The function *f* is not onto because there is no integer x with $x^2 = -1$, for instance.

	Injective	Surjective	
1	❌	❌	Possible
2	✅	❌	Possible
3	❌	✅	Possible
4	✅	✅	Possible

One-to-one Correspondence [Bijection]

The function ***f*** is a ***one-to-one correspondence***, or a ***bijection***, if it is both ***one-to-one*** and ***onto***. We also say that such a function is bijective.

EXAMPLE: Let ***f*** be the function from **{a, b, c, d}** to **{1, 2, 3, 4}** with ***f(a) = 4, f(b) = 2, f(c) = 1,*** and ***f(d) = 3***. Is ***f*** a bijection?
SOLUTION: The function ***f*** is one-to-one and onto. It is one-to-one because no two values in the domain are assigned the same function value. It is onto because all four elements of the codomain are images of elements in the domain. Hence, ***f*** is a bijection.

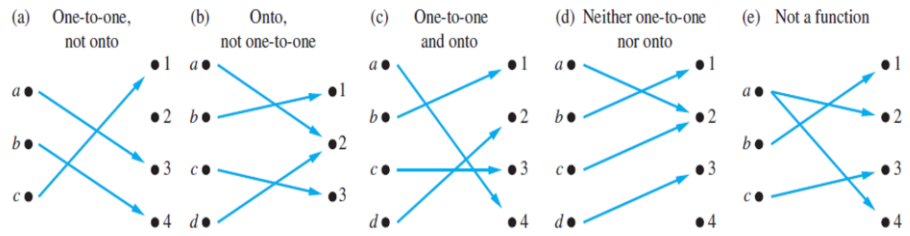


Fig: Examples of different types of correspondences.

- **Bijection = Injection + Surjection**
- **Bijection** কে Invertible function ও বলা হয় কেননা Bijection কে inverse করা যায়।

Techniques

Suppose that $f : A \to B$.

- ***To show that f is injective:*** Show that if $f(x) = f(y)$ for arbitrary $x, y \in A$, then $x = y$.
- ***To show that f is not injective:*** Find particular elements $x, y \in A$ such that $x \neq y$ and $f(x) = f(y)$.
- ***To show that f is surjective:*** Consider an arbitrary element $y \in B$ and find an element $x \in A$ such that $f(x) = y$.
- ***To show that f is not surjective:*** Find a particular $y \in B$ such that $f(x) \neq y$ for all $x \in A$.

Cardinality of Sets

The sets ***A*** and ***B*** have the ***same cardinality*** if and only if there is a ***one-to-one correspondence*** from ***A*** to ***B***. When ***A*** and ***B*** have the same cardinality, we write $|A| = |B|$.

In another words, if there is a one-to-one function from ***A*** to ***B***, the cardinality of ***A*** is ***less than or the same*** as the cardinality of ***B*** and we write $|A| \leq |B|$. Moreover, when $|A| \leq |B|$ and ***A*** and ***B*** have ***different*** cardinality, we say that the cardinality of ***A*** is ***less than*** the cardinality of ***B*** and we write $|A| < |B|$.

Countable and Uncountable Sets

A set that is either ***finite*** or has the ***same cardinality*** as ***the set of positive integers*** is called ***countable***. A set that is ***not countable*** is called ***uncountable***.

When an ***infinite*** set ***S*** is ***countable***, we denote the cardinality of ***S*** by

$\aleph_0$ (where $\aleph$ is aleph, the first letter of the Hebrew alphabet). We write $|S| = \aleph_0$ and say that S has cardinality ***"aleph null."***

EXAMPLE: Show that the set of odd positive integers is a countable set.

SOLUTION: To show that the set of odd positive integers is countable, we will exhibit a one-to-one correspondence between this set and the set of positive integers. Consider the function $f(n) = 2n-1$, from Z^+ to the set of odd positive integers.

We show that f is a ***one-to-one correspondence*** by showing that it is ***both one-to-one and onto***. To see that it is one-to-one, suppose that $f(n) = f(m)$. Then $2n-1 = 2m-1$, so $n = m$. To see that it is onto, suppose that t is an odd positive integer. Then t is 1 less than an even integer $2k$, where k is a natural number. Hence $t = 2k-1 = f(k)$.

Inverse Functions

Let ***f*** be a one-to-one correspondence from the set ***A*** to the set ***B***. The ***inverse function of f*** is the function that assigns to an element ***b*** belonging to ***B*** to the unique element ***a*** in ***A*** such that ***f(a) = b***. The inverse function of ***f*** is denoted by f^{-1}. Hence, $f^{-1}(b) = a$ when $f(a) = b$.

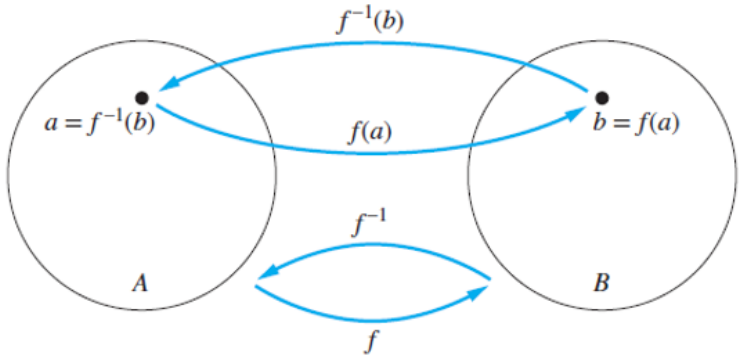


Fig : The function f^{-1} is the inverse function of f

A **one-to-one correspondence** is called **invertible** because we can **define** an inverse of this function. A function is **not invertible** if it is **not a one-to-one correspondence**, because the inverse of such a function **does not exist**.

EXAMPLE: Let ***f*** be the function from ***{a, b, c}*** to ***{1, 2, 3}*** such that ***f(a) = 2***, ***f(b) = 3***, and ***f(c) = 1***. Is ***f*** invertible, and if it is, what is its inverse?

SOLUTION: The function f is invertible because it is a one-to-one correspondence. The inverse function f^{-1} reverses the correspondence given by f, so $f^{-1}(1) = c$, $f^{-1}(2) = a$, and $f^{-1}(3) = b$.

EXAMPLE: Let ***f*** be the function from ***R*** to ***R*** with $f(x) = x^2$. Is f invertible?

SOLUTION: Because ***f(−2) = f(2) = 4***, ***f*** is not one-to-one. If an inverse function were defined, it would have to assign ***two*** elements to ***4***. Hence, ***f*** is not invertible. (Note we can also show that ***f*** is not invertible because it is not onto.)

Compositions of Functions

Let ***g*** be a function from the set ***A*** to the set ***B*** and let ***f*** be a function from the set ***B*** to the set ***C***. The composition of the functions ***f*** and g, denoted for all ***a ∈ A*** by ***f∘g***, is the function from ***A*** to ***C*** defined by ***(f∘g)(a) = f(g(a))***.

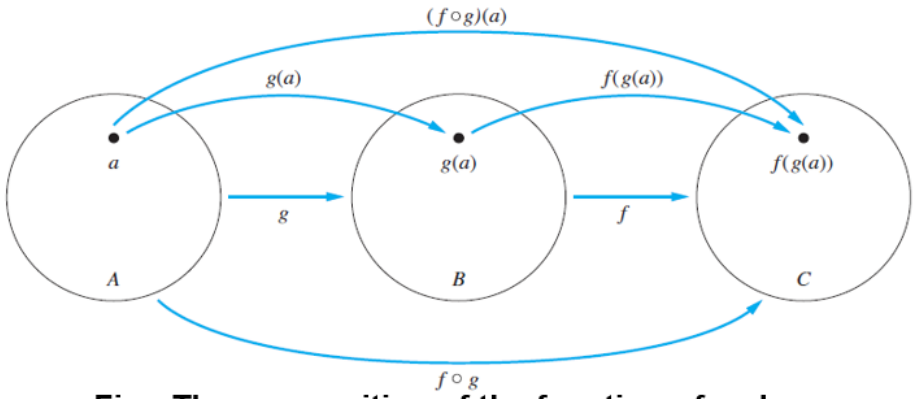


Fig : The composition of the functions *f* and *g*

EXAMPLE: Let ***g*** be the function from the set ***{a, b, c}*** to ***itself*** such that ***g(a) = b, g(b) = c, and g(c) = a.*** Let ***f*** be the function from the set ***{a, b, c}*** to the set ***{1, 2, 3}*** such that ***f(a) = 3, f(b) = 2, and f(c) = 1***. What is the composition of ***f*** and ***g***, and what is the composition of ***g*** and ***f*** ?

SOLUTION: The composition *f∘g* is defined by *(f∘g)(a) = f(g(a)) = f(b) = 2*, *(f∘g) (b) = f(g(b)) = f(c) = 1*, and *(f∘g)(c) = f(g(c)) = f(a) = 3*.
Note that g∘f is not defined, because the range of f is not a subset of the domain of g.

EXAMPLE: Let ***f*** and ***g*** be the functions defined by ***f : R → R⁺∪ {0}*** with ***f(x) = x²*** and ***g : R⁺∪ {0} → R*** with ***g(x) = √x*** (where ***√x*** is the

nonnegative square root of ***x***). What is the function ***(f∘g)(x)***?

SOLUTION: The domain of ***(f∘g)(x) = f(g(x))*** is the domain of ***g***, which is ***$R^+ \cup \{0\}$***, the set of nonnegative real numbers. If ***x*** is a ***nonnegative real number***, we have ***$(f\circ g)(x) = f(g(x)) = f(\sqrt{x}) = (\sqrt{x})^2 = x$***. The range of ***f∘g*** is the image of the range of g with respect to the function ***f***. This is the set ***$R^+ \cup \{0\}$***, the set of nonnegative real numbers. Summarizing, ***$f : R^+ \cup \{0\} \rightarrow R^+ \cup \{0\}$*** and ***f(g(x)) = x*** for all ***x***.

Explanation

When the composition of a function and its inverse is formed, in either order, an identity function is obtained. To see this, suppose that ***f*** is a ***one-to-one correspondence*** from the set ***A*** to the set ***B***. Then the inverse function ***f^{-1}*** exists and is a ***one-to-one correspondence*** from ***B*** to ***A***. The inverse function reverses the correspondence of the original function, so ***$f^{-1}(b) = a$*** when ***f(a) = b***, and ***f(a) = b*** when ***$f^{-1}(b) = a$***. Hence, ***$(f^{-1}\circ f)(a) = f^{-1}(f(a)) = f^{-1}(b) = a$***, and ***$(f\circ f^{-1})(b) = f(f^{-1}(b)) = f(a) = b$***. Consequently ***$f^{-1}\circ f = I_A$*** and ***$f \circ f^{-1} = I_B$,*** where ***$I_A$*** and ***$I_B$*** are the identity functions on the sets ***A*** and ***B***, respectively. That is, ***$(f^{-1})^{-1} = f$*** .

Some Important Functions

The ***floor function*** assigns to the real number ***x*** the ***largest*** integer that is ***less than or equal to x***. The value of the floor function at ***x*** is denoted by ***⌊x⌋***.

The ***ceiling function*** assigns to the real number ***x*** the ***smallest*** integer that is ***greater than or equal to x***. The value of the ceiling function at ***x*** is denoted by ***⌈x⌉***.

TABLE 1 **Useful Properties of the Floor and Ceiling Functions.** (n is an integer, x is a real number)
(1a) $\lfloor x \rfloor = n$ if and only if $n \le x < n + 1$ (1b) $\lceil x \rceil = n$ if and only if $n - 1 < x \le n$ (1c) $\lfloor x \rfloor = n$ if and only if $x - 1 < n \le x$ (1d) $\lceil x \rceil = n$ if and only if $x \le n < x + 1$
(2) $x - 1 < \lfloor x \rfloor \le x \le \lceil x \rceil < x + 1$
(3a) $\lfloor -x \rfloor = -\lceil x \rceil$ (3b) $\lceil -x \rceil = -\lfloor x \rfloor$
(4a) $\lfloor x + n \rfloor = \lfloor x \rfloor + n$ (4b) $\lceil x + n \rceil = \lceil x \rceil + n$

EXAMPLE: Prove that if ***x*** is a ***real number***, then ***⌊2x⌋ = ⌊x⌋ + ⌊x + 1⁄2⌋***.

SOLUTION: To prove this statement we let ***x = n + ϵ***, where ***n*** is an integer and ***0 ≤ ϵ < 1***. There Examples are two cases to consider, depending on whether ***ϵ*** is l***ess than, or greater than or equal to 1⁄2*** . (The reason we choose these two cases will be made clear in the proof.)

We first consider the case when ***0 ≤ ϵ < 1⁄2*** . In this case, ***2x = 2n + 2ϵ*** and ***⌊2x⌋ = 2n*** because ***0 ≤ 2ϵ < 1***. Similarly, ***x + 1⁄2 = n + (1⁄2 + ϵ)***, so ***⌊x + 1⁄2 ⌋ = n***, because ***0 < 1⁄2 + ϵ < 1***. Consequently, ***⌊2x⌋ = 2n*** and ***⌊x⌋ + ⌊x + 1⁄2 ⌋ = n + n = 2n.***

Next, we consider the case when ***1⁄2 ≤ ϵ < 1***. In this case, ***2x = 2n + 2ϵ = (2n + 1) + (2ϵ − 1)***. Because ***0 ≤ 2ϵ − 1 < 1***, it follows that ***⌊2x⌋ = 2n + 1***. Because ***⌊x + 1⁄2⌋ = ⌊n + (1⁄2 + ϵ)⌋ = ⌊n + 1 + (ϵ − 1⁄2)⌋*** and ***0 ≤ ϵ − 1⁄2 < 1***, it follows that ***⌊x + 1⁄2⌋ = n + 1.*** Consequently, ***⌊2x⌋ = 2n + 1 and ⌊x⌋ + ⌊x + 1⁄2⌋ = n + (n + 1) = 2n + 1.*** This concludes the proof.

Sequences and Summations

Sequences

Definition: A sequence is a function from a subset of the set of integers (usually either the set {0, 1, 2, …} or the set {1, 2, 3, …}) to a set S. We use the notation a_n to denote the image of the integer n. We call a_n a term of the sequence.

Example: Consider the sequence $\{a_n\}$, where

$a_n = 1/n$.

The list of the terms of this sequence, beginning with a_1, namely,

$a_1, a_2, a_3, a_4, \ldots$,

starts with 1, 1/2, 1/3, 1/4, . …

Sequences: Geometric Progression

Definition: A geometric progression is a sequence of the form

$a, ar, ar^2, \ldots, ar^n, \ldots$

where the initial term *a* and the common ratio *r* are real numbers.

Remark: A geometric progression is a discrete analogue of the exponential function $f(x) = ar^x$.

Example: The sequences $\{b_n\}$ with $b_n = (-1)^n$, $\{c_n\}$ with $c_n = 2 \cdot 5^n$, and $\{d_n\}$ with $d_n = 6 \cdot (1/3)^n$ are geometric progressions with initial term and common ratio equal to 1 and −1; 2 and 5; and 6 and 1/3, respectively, if we start at n = 0. The list of terms $b_0, b_1, b_2, b_3, b_4, \ldots$ begins with

1, −1, 1, −1, 1,...;

the list of terms $c_0, c_1, c_2, c_3, c_4, \ldots$ begins with

2, 10, 50, 250, 1250,...;

and the list of terms $d_0, d_1, d_2, d_3, d_4, \ldots$ begins with

6, 2, 2/3, 2/9, 2/27 ,....

Sequences: Arithmetic Progression

Definition: An Arithmetic Progression is a sequence of the form
a, a+d, a+2d, ..., a+nd, ...

where the initial term *a* and the common difference *d* are real numbers.

Example: The sequences $\{s_n\}$ with $s_n = -1 + 4n$ and $\{t_n\}$ with $t_n = 7 - 3n$ are both arithmetic progressions with initial terms and common differences equal to −1 and 4, and 7 and −3, respectively, if we start at n = 0. The list of terms $s_0, s_1, s_2, s_3, \ldots$ begins with
$-1, 3, 7, 11, \ldots$,
and the list of terms $t_0, t_1, t_2, t_3,, \ldots$ begins with
$7, 4, 1, -2, \ldots$.
Sequences of the form $a_1, a_2, \ldots, a_n$ are often used in computer science. These finite sequences are also called *strings*. This string is also denoted by $a_1\, a_2 \ldots a_n$. The length of a string is the number of terms in this string. The empty string, denoted by λ, is the string that has no terms. The *empty string* has length zero. For example, The string abcd is a string of length four.

Recurrence Relations

Definition: A recurrence relation for the sequence $\{a_n\}$ is an equation that expresses a_n in terms of one or more of the previous terms of the sequence, namely, $a_0, a_1, \ldots, a_{n-1}$, for all integers n with $n \geq n_0$, where n_0 is a nonnegative integer. A sequence is called a solution of a recurrence relation if its terms satisfy the recurrence relation. (A recurrence relation is said to recursively define a sequence.)
Example: Let $\{a_n\}$ be a sequence that satisfies the recurrence relation $a_n = a_{n-1} + 3$ for n = 1, 2, 3, … , and suppose that $a_0 = 2$. What are a_1,

a_2, and a_3?

Solution: We see from the recurrence relation that $a_1 = a_0 + 3 = 2 + 3 = 5$.

It then follows that $a_2 = 5 + 3 = 8$

and $a_3 = 8 + 3 = 11$.

Recurrence Relations: Fibonacci sequence

Definition: The Fibonacci sequence, f_0, f_1, f_2, … , is defined by the initial conditions $f_0 = 0$, $f_1 = 1$, and the recurrence relation

$f_n = f_{n-1} + f_{n-2}$

for n = 2, 3, 4, … .

Example: Find the Fibonacci numbers f_2, f_3, f_4, f_5, and f_6.

Solution: The recurrence relation for the Fibonacci sequence tells us that we find successive terms by adding the previous two terms. Because the initial conditions tell us that $f_0 = 0$ and $f_1 = 1$, using the recurrence relation in the definition we find that,

$f_2 = f_1 + f_0 = 1 + 0 = 1$

$f_3 = f_2 + f_1 = 1 + 1 = 2$

$f_4 = f_3 + f_2 = 2 + 1 = 3$

$f_5 = f_4 + f_3 = 3 + 2 = 5$

$f_6 = f_5 + f_4 = 5 + 3 = 8$

Recurrence Relations: Compound Interest

Example: Suppose that a person deposits $10,000 in a savings account at a bank yielding 11% per year with interest compounded annually. How much will be in the account after 30 years?

Solution: To solve this problem, let P_n denote the amount in the account after *n* years. Because the amount in the account after *n*

years equals the amount in the account after *n−1* years plus interest for the *n*th year, we see that the sequence $\{P_n\}$ satisfies the recurrence relation

$P_n = P_{n-1} + 0.11P_{n-1} = (1.11)P_{n-1}$.

The initial condition is $P_0 = 10{,}000$. We can use an iterative approach to find a formula for P_n . Note that

$P_1 = (1.11)P_0$

$P_2 = (1.11)P_1 = (1.11)^2P_0$

$P_3 = (1.11)P_2 = (1.11)^3P_0$

.

.

.

$P_n = (1.11)P_{n-1} = (1.11)^nP_0$.

When we insert the initial condition $P_0 = 10{,}000$, the formula $P_n = (1.11)^n10{,}000$ is obtained.

Inserting n = 30 into the formula $P_n = (1.11)^n10{,}000$ shows that after 30 years the account contains

$P_{30} = (1.11)^{30}10{,}000 = \$228{,}922.97$.

Special Integer Sequences

Example 1: Find formulae for the sequences with the following first five terms: (a) 1, 1/2, 1/4, 1/8, 1/16 (b) 1, 3, 5, 7, 9 (c) 1, −1, 1, −1, 1.

Solution:

(a) We recognize that the denominators are powers of 2. The sequence with $a_n = 1/2^n$, n = 0, 1, 2, … is a possible match. This proposed sequence is a geometric progression with a = 1 and r = 1/2.

(b) We note that each term is obtained by adding 2 to the previous term. The sequence with $a_n = 2n + 1$, n = 0, 1, 2, … is a possible

match. This proposed sequence is an arithmetic progression with a = 1 and d = 2.

(c) The terms alternate between 1 and −1. The sequence with $a_n = (-1)^n$, n = 0, 1, 2… is a possible match. This proposed sequence is a geometric progression with a = 1 and r = −1.

Example 2: How can we produce the terms of a sequence if the first 10 terms are 1, 2, 2, 3, 3, 3, 4, 4, 4, 4?

Solution: In this sequence, the integer 1 appears once, the integer 2 appears twice, the integer 3 appears three times, and the integer 4 appears four times. A reasonable rule for generating this sequence is that the integer *n* appears exactly *n* times, so the next five terms of the sequence would all be 5, the following six terms would all be 6, and so on. The sequence generated this way is a possible match.

Summations

Next, we consider the addition of the terms of a sequence. For this we introduce summation notation. We begin by describing the notation used to express the sum of the terms

$a_m, a_{m+1}, \ldots, a_n$

from the sequence $\{a_n\}$. We use the notation

$$\sum_{j=m}^{n} a_j, \qquad \sum_{j=m}^{n} a_j, \qquad \text{or} \qquad \sum_{m \le j \le n} a_j$$

(read as the sum from j = m to j = n of a_j) to represent

$a_m + a_{m+1} + \ldots + a_n$

Here, the variable *j* is called the index of summation, and the choice of the letter *j* as the variable is arbitrary; that is, we could have used any other letter, such as *i* or *k*. Or, in notation,

$$\sum_{j=m}^{n} a_j = \sum_{i=m}^{n} a_i = \sum_{k=m}^{n} a_k.$$

Sum of terms of a geometric progression

Theorem:

If a and r are real numbers and $r \neq 0$, then

$$\sum_{j=0}^{n} ar^j = \begin{cases} \dfrac{ar^{n+1} - a}{r - 1} & \text{if } r \neq 1 \\ (n+1)a & \text{if } r = 1. \end{cases}$$

Proof: Let $S_n = \sum_{j=0}^{n} ar^j.$

To compute S, first multiply both sides of the equality by r and then manipulate the resulting sum as follows:

$$\begin{aligned}
rS_n &= r\sum_{j=0}^{n} ar^j && \text{substituting summation formula for } S \\
&= \sum_{j=0}^{n} ar^{j+1} && \text{by the distributive property} \\
&= \sum_{k=1}^{n+1} ar^k && \text{shifting the index of summation, with } k = j + 1 \\
&= \left(\sum_{k=0}^{n} ar^k\right) + (ar^{n+1} - a) && \text{removing } k = n + 1 \text{ term and adding } k = 0 \text{ term} \\
&= S_n + (ar^{n+1} - a) && \text{substituting } S \text{ for summation formula}
\end{aligned}$$

From these equalities, we see that

$$rS_n = S_n + (ar^{n+1} - a).$$

Solving for S_n shows that if $r \neq 1$, then

$$S_n = \frac{ar^{n+1} - a}{r - 1}.$$

If $r = 1$, then the $S_n = \sum_{j=0}^{n} ar^j = \sum_{j=0}^{n} a = (n + 1)a.$

সহজভাবে প্রমাণ করলে,

$Sn = a + ar + ar^2 + \ldots\ldots\ldots\ldots + ar^{n-1}$

$rSn = ar + ar^2 + ar^3 + \ldots\ldots\ldots\ldots + ar^n$

এভাবে $r = 0$ এর জন্য।

যেহেতু, $r \neq 0$,

$Sn = a + ar + ar^2 + \ldots\ldots\ldots\ldots + ar^n$ ---------- (i)

$rSn = ar + ar^2 + ar^3 + \ldots\ldots\ldots\ldots + ar^{n+1}$ ---------- (ii)

(ii) – (i)

$rSn – Sn = ar^{n+1} – a$

$\Rightarrow Sn (r – 1) = ar^{n+1} – a$

$\Rightarrow Sn = ar^{n+1} – a / r – 1$ [$r \neq 1$]

Mathematical Induction

5.1.1 Introduction

Suppose that we have an infinite ladder, as shown in Figure 1, and we want to know whether we can reach every step on this ladder. We know two things:

1. We can reach the first rung of the ladder.
2. If we can reach a particular rung of the ladder, then we can reach the next rung.

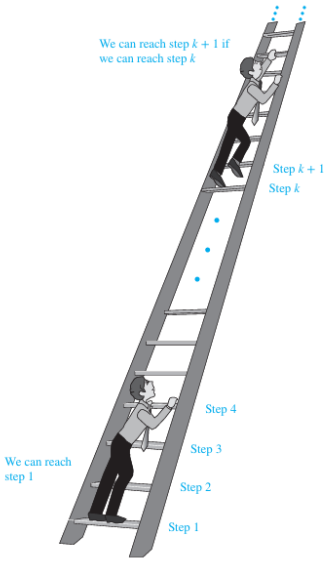


FIGURE 1 **Climbing an infinite ladder.**

For example, after 100 uses of (2), we know that we can reach the 101st rung. But can we conclude that we are able to reach every rung of this infinite ladder? The answer is yes, something we can verify using an important proof technique called mathematical induction. That is, we can show that the statement that we can each the nth rung of the ladder is true for all positive integers n.

Mathematical induction is an extremely important proof technique that can be used to prove assertions of this type.

PRINCIPLE OF MATHEMATICAL INDUCTION To prove that P(n) is true for all positive integers n, where P(n) is a propositional function, we

complete two steps:

BASIS STEP: We verify that P(1) is true.

INDUCTIVE STEP: We show that the conditional statement P (k) → P (k + 1) is true for all positive integers k.

$P(n) = 1 + 2 + 3 + \ldots\ldots\ldots + n = \frac{n(n+1)}{2}$

Implication: If P(k) is true, P(k+1) is also true.

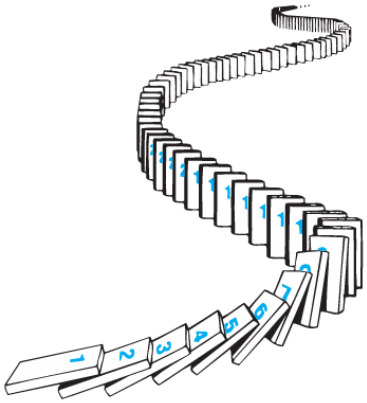


FIGURE 2 Illustrating how mathematical induction works using dominoes.

$\forall k\,(P(k) \rightarrow P(k+1))$

EXAMPLE 1 Show that if n is a positive integer, then $1 + 2 + \cdots + n = \frac{n(n+1)}{2}$. ["∀n P(n)" is True এটা প্রমাণ করতে বলেছে। মূল উদ্দেশ্য এটাই।]

Solution: Let P(n) be the proposition/Predicate that the sum of the first n positive integers, $1 + 2 + \cdots n$ is $\frac{n(n+1)}{2}$

BASIS STEP:

We need to prove, P(1) is true.

L.H.S = 1

$R.H.S = \frac{1(1+1)}{2} = 1$

As, L.H.S = R.H.S. So, P(1) is True.

INDUCTIVE STEP:

$\forall k(P(k) \rightarrow P(k+1))$

We assume, P(k) is true.

$P(k) = 1 + 2 + 3 + ... + k = \frac{k(k+1)}{2}$

We have to prove, P(k+1) is true.

$P(k+1) = 1 + 2 + 3 + ... + k + (k+1) = \frac{(k+1)(k+2)}{2}$

$L.H.S = 1 + 2 + 3 + ... + k + (k+1)$

$= \frac{k(k+1)}{2} + (k+1)$

$= \frac{k(k+1)+2(k+1)}{2}$

$= \frac{(k+1)(k+2)}{2}$

$= R.H.S$

Therefore, P(k+1) is True.

Final Statement এ এভাবে লিখবো,

We have shown, P(1) is true and we have also shown $\forall k(P(k) \rightarrow P(k+1))$ is true. Using both proof/statement we can say that "$\forall n$ P(n)"is true.

(Proved)

Proving Inequalities

$P(n) = n < 2^n$; $n \in$ Non-Negative integers [0 থেকে শুরু].

Solution:

Basis Step:

We need to prove P(0) is true.

L.H.S = 0

$R.H.S = 2^0 = 1$

L.H.S < R.H.S. Therefore, P(0) is true.

Inductive Step:

$\forall k(P(k) \rightarrow P(k+1))$

We assume, P(k) is true.

$P(k) = k < 2^k$

We have to prove, P(k+1) is true.

$P(k+1) = k+1 < 2^{k+1}$

L.H.S = k+1

R.H.S = 2^k+1

$= 2.2^k$

$= 2^k+2^k$

Or,

L.H.S = $k+1 < 2^k+1$

L.H.S $< 2^k+1 \leq 2^k+2^k$

L.H.S $< 2^k+2^k$

L.H.S $< 2^{k+1}$

L.H.S < R.H.S $\forall n\ P(n)$

Therefore, P(k+1) is True.

We have shown, P(0) is true and we have also shown ∀k(P(k) → P(k+1)) is true. Using both proof/statement we can say that "∀n P(n)"is true.

(Proved)

Proving Divisibility

Prove P(n) = n^3-n is divisible by 3. For all Positive integers.

Solution:

Basis:

P(1) = 1^3 - 1 = 0, which is divisible by 3.

Therefore, P(0) is true.

Inductive:

∀k(P(k) → P(k +1))

We assume, P(k) is true.

P(k) = k^3 – k is divisible by 3.

We have to prove, P(k+1) is true.

P(k+1) = $(k+1)^3 – (k+1)$

= $(k+1) \{(k+1)^2 – 1\}$

= $(k+1) (k^2+2k)$

= $k^3+2k^2+k^2+2k$

= k^3-k+3k^2+3k

= $k^3-k+3(k^2+k)$

Therefore, P(k+1) is true.

We have shown, P(1) is true and we have also shown ∀k(P(k) → P(k+1)) is true. Using both proof/statement we can say that "∀n P(n)"is true.

(Proved)

Divisibility and Modular Arithmetic

$a \mid b = \frac{b}{a}$

Theorem-1: Let a, b & c be integers, where $a \neq 0$. Then if $a \mid b$ and $a \mid c$, then $a \mid (b+c)$.

Proof:

$b = ma$

$c = na$ [m, n হচ্ছে multiplicand বা গুণনীয়ক।]

$a \mid (b+c)$

$= \frac{b+c}{a}$

$= \frac{ma+na}{a}$

$= \frac{a\,(m+n)}{a}$

The Division Algorithm

If a is an integer & d is a positive integer then there exists unique q and r with $0 \leq r < d$, such that $a = dq + r$

Quotient, q = a div d (a/d) ; Remainder, r = a mod d (a%d).

$52 = 5 \times 10 + 2$

Here, 52 = a; 5 = d; 10 = q; 2 = r.